

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO4: Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 10

Total Marks:100

Annexure A

Scenario-Based Assessment

Code of Conduct:

*I (Shaik Mohammed Abrar/192472185) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Shaik Mohammed Abrar

RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

NOTE:

For every question, answer the following:

- Identify the numerical and binary features used for KNN classification.
- Calculate the **Euclidean distance** between the new sample and all training samples
- Calculate the **Manhattan distance** between the new sample and all training samples.
- Calculate the **Minkowski distance** with $p = 3$ between the new sample and all training samples.
- Calculate the **Hamming distance** between the new sample and all training samples using binary features.
- Identify the **3 nearest neighbours** separately for each distance measure.
- Predict the class label using majority voting for each distance measure.
- Compare the results and explain which distance measure is most suitable for the given scenario.

Total = 100 Marks

Question 1: Student Pass or Fail Prediction

A college wants to predict whether a student will Pass or Fail based on academic and activity-related details.

Student	Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done	Result
S1	2	60	0	0	0	Fail
S2	3	65	1	0	0	Fail
S3	6	80	1	1	1	Pass
S4	7	85	1	1	1	Pass
S5	4	70	1	0	1	Fail
S6	8	90	1	1	1	Pass

A new student has the following details:

Study Hours	Attendance %	Assignment Submitted	Lab Participation	Revision Done
5	75	1	1	0

Using KNN with $K = 3$, classify the new student as Pass or Fail using Euclidean, Manhattan, Minkowski, and Hamming distance.

COT-NT1

(1) Student Pass or Fail Prediction ← Question

New student

$$X = (5, 75, 1, 1, 0)$$

Euclidean Distance

$$\text{Formula: } D_L(X, Y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

For S1

$$S1 : (2, 60, 0, 0, 0)$$

New student:-

$$(5, 75, 1, 1, 0)$$

$$\begin{aligned} D(X, S1) &= \sqrt{(5-2)^2 + (75-60)^2 + (1-0)^2 + (1-0)^2 + (0-0)^2} \\ &= \sqrt{3^2 + 15^2 + 1^2 + 1^2 + 0^2} \\ &= \sqrt{9 + 225 + 1 + 1} \\ &= \sqrt{236} \end{aligned}$$

$$D(X, S1) = 15.3623$$

all Euclidean Distances :-

Student	Calculation	Distance
S1	$\sqrt{3^2 + 15^2 + 1^2 + 1^2 + 0^2}$	15.3623
S2	$\sqrt{2^2 + 10^2 + 0^2 + 1^2 + 0^2}$	10.2470
S3	$\sqrt{(-1)^2 + (-1)^2 + 0^2 + 10^2 + (-1)^2}$	10.2470 5.2915
S4	$\sqrt{(-2)^2 + (-10)^2 + 0^2 + 0^2 + (-1)^2}$	10.2470
S5	$\sqrt{1^2 + 5^2 + 0^2 + 1^2 + (-1)^2}$	5.2915
S6	$\sqrt{(-3)^2 + (-15)^2 + 0^2 + 0^2 + (-1)^2}$	15.3247

Nearest Neighbour :

$S_3 \rightarrow \text{Pass}$, $S_5 \rightarrow \text{Fail}$

$S_3/S_4 \rightarrow \text{tie}$

S_2 comes before S_4

S_3, S_5, S_2

classes: Pass, Fail, Fail

Majority :- Fail

Manhattan Distance

Formula : $D_M(X, Y) = \sum_{i=1}^n |x_i - y_i|$

for S_1 :- $D(X, S_1) = |5-2| + |25-60| + |1-0| + |1-0| + |0-0|$
 $= 3+15+1+1+0$

$D(X, S_1) = 20$

All Manhattan Distances

Student	Calculation	Distance
S_1	$3+15+1+1+0$	20
S_2	$2+10+0+1+0$	13
S_3	$1+5+0+0+1$	7
S_4	$2+10+0+0+1$	13
S_5	$1+5+0+1+1$	8
S_6	$3+15+0+0+1$	19

3 Nearest Neighbours : S_3, S_5, S_2

Classes :-

$S_3 \rightarrow \text{Pass}$

$S_5 \rightarrow \text{Fail}$

$S_2 \rightarrow \text{Fail}$

$\therefore$ Prediction = Fail

Hamming Distance

Binary Features: (Assignment, Lab, Revision)

$$X = (1, 1, 0)$$

Formula:-

$$D_H(X, Y) = \sum_{i=1}^n I(x_i \neq y_i)$$

$$\text{where } I(x_i \neq y_i) = \begin{cases} 1 & \text{if different} \\ 0 & \text{if same} \end{cases}$$

S_j :- Binary Features: (0, 0, 0)

$$X = (1, 1, 0)$$

Comparison:- $(1 \neq 0), (1 \neq 0), (0 = 0)$

$$\therefore D_H = 1 + 1 + 0$$

$$D_H(S_1) = 2$$

<u>Student</u>	<u>Binary Features</u>	<u>Hamming Distance</u>
S_1	(0, 0, 0)	2
S_2	(1, 0, 0)	1
S_3	(1, 1, 1)	1
S_4	(1, 1, 1)	1
S_5	(0, 1)	2
S_6	(1, 1, 1)	1

Nearest distance:-

$$S_2, S_3, S_4, S_6$$

$$D_H = 1$$

Since $k=3$, a tie-breaking rule is required

$$S_2, S_3, S_4$$

Class:- $S_2 \rightarrow \text{Fail}, S_3 \rightarrow \text{Pass}, S_4 \rightarrow \text{Pass}$

$$\therefore \text{Prediction} = \text{Pass}$$

Minkowski Distance, $p=3$

Formula

$$D_m(X, Y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

$$p=3 \therefore D(X, Y) = \left(\sum_{i=1}^n |x_i - y_i|^3 \right)^{1/3}$$

for S_1

$$D(X, S_1) = [(5-2)^3 + (35-40)^3 + (1-0)^3 + (1-0)^3 + (0+0)^3]^{1/3}$$

$$= [3^3 + 15^3 + 1^3 + 1^3 + 0^3]^{1/3}$$

$$= [27 + 3375 + 1 + 1]^{1/3}$$

$$= 3404^{1/3}$$

$$D(X, S_1) = 15.0428$$

all Minkowski Distance

Student

Distance

S_1

15.0428

S_2

10.0299

S_3

5.0265

S_4

10.0299

S_5

5.0399

S_6

15.0414

3-Nearest Neighbours

S_3, S_5, S_2

Class

Pass, Fail, Fail

$\therefore$ Prediction: Fail

Output:

STUDENT DATA

	Student	Study_Hours	Attendance	Assignment_Submitted	Lab_Participation
0	S1	2	60	0	0
1	S2	3	65	1	0
2	S3	6	80	1	1
3	S4	7	85	1	1
4	S5	4	70	1	0
5	S6	8	90	1	1

	Revision_Done	Result
0	0	Fail
1	0	Fail
2	1	Pass
3	1	Pass
4	1	Fail
5	1	Pass

DISTANCE CALCULATIONS

Student	Result	Euclidean	Manhattan	Minkowski	Hamming
S1	Fail	15.3623	20	15.0428	2
S2	Fail	10.2470	13	10.0299	1
S3	Pass	5.1962	7	5.0265	1
S4	Pass	10.2470	13	10.0299	1
S5	Fail	5.2915	8	5.0397	2
S6	Pass	15.3297	19	15.0414	1

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Student	Result	Distance
S3	Pass	5.196152
S5	Fail	5.291503
S4	Pass	10.246951

Majority Voting:

Result

Pass 2

Fail 1

Name: count, dtype: int64

Prediction: Pass

MANHATTAN DISTANCE

3 Nearest Neighbours:

Student	Result	Distance
S3	Pass	7
S5	Fail	8
S4	Pass	13

Majority Voting:

Result

Pass 2
Fail 1
Name: count, dtype: int64

Prediction: Pass

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MINKOWSKI DISTANCE (p = 3)

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3 Nearest Neighbours:

Student	Result	Distance
S3	Pass	5.026526
S5	Fail	5.039684
S4	Pass	10.029910

Majority Voting:

Result

Pass 2
Fail 1
Name: count, dtype: int64

Prediction: Pass

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HAMMING DISTANCE

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3 Nearest Neighbours:

Student	Result	Distance
S2	Fail	1
S3	Pass	1
S4	Pass	1

Majority Voting:

Result

Pass 2
Fail 1
Name: count, dtype: int64

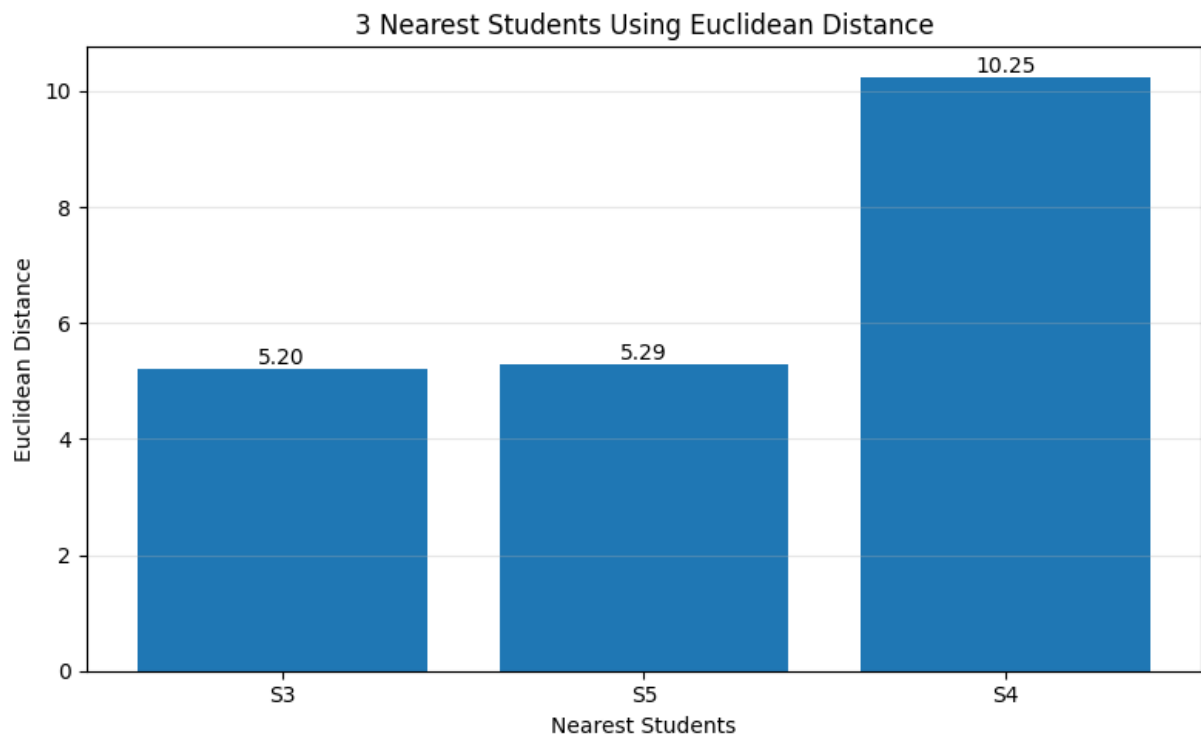
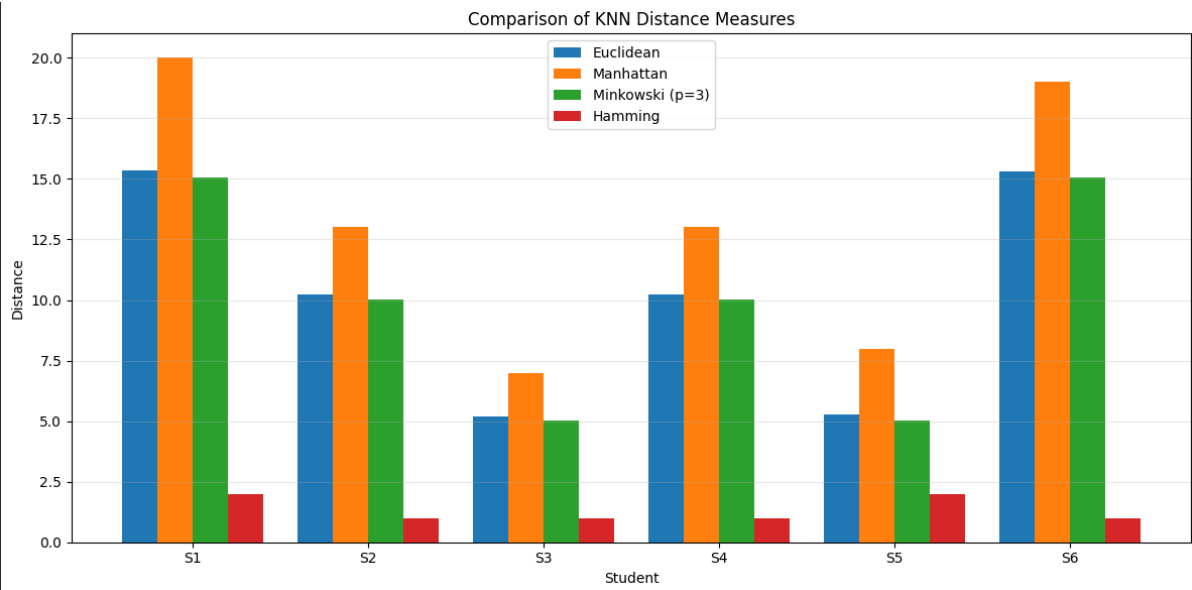
Prediction: Pass

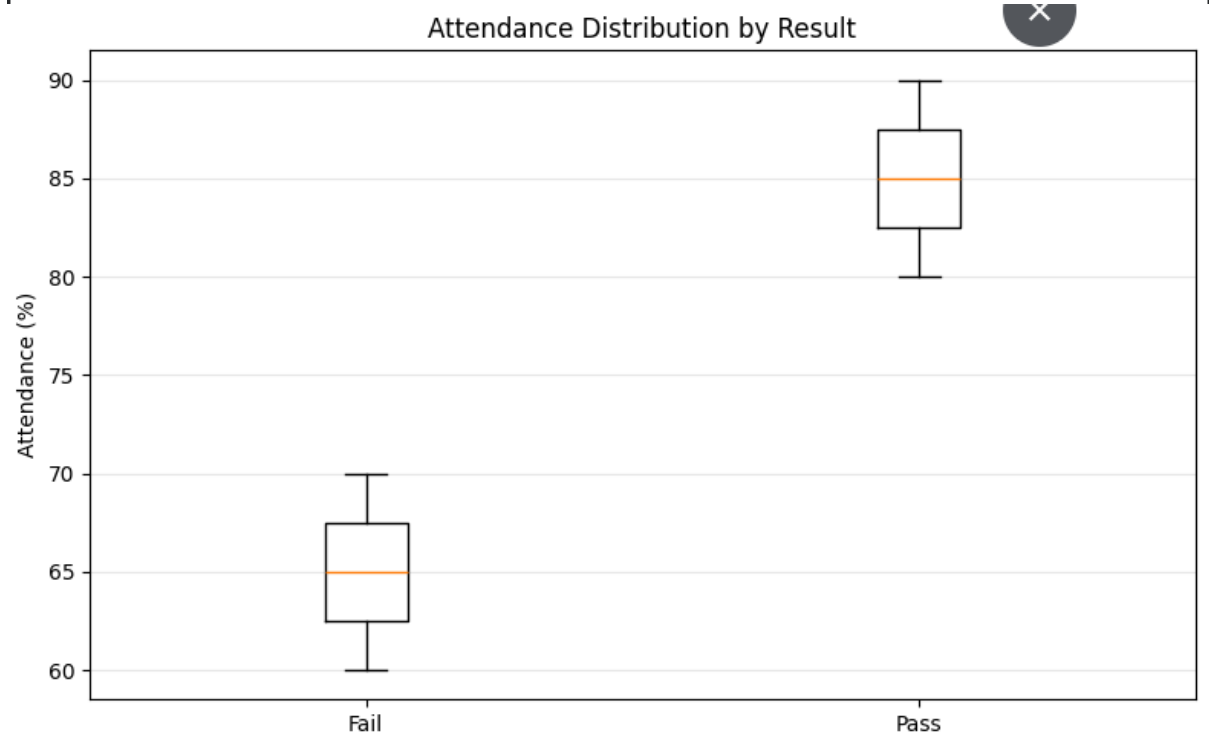
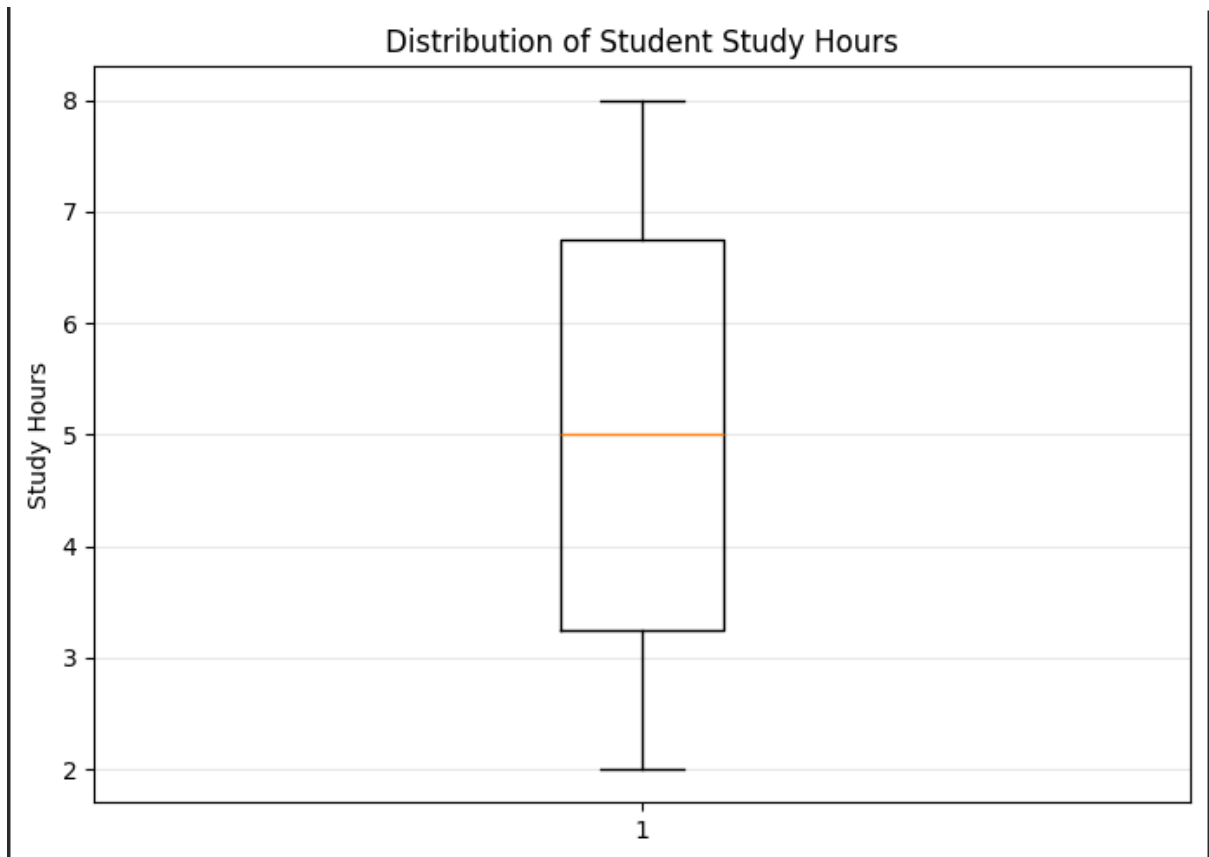
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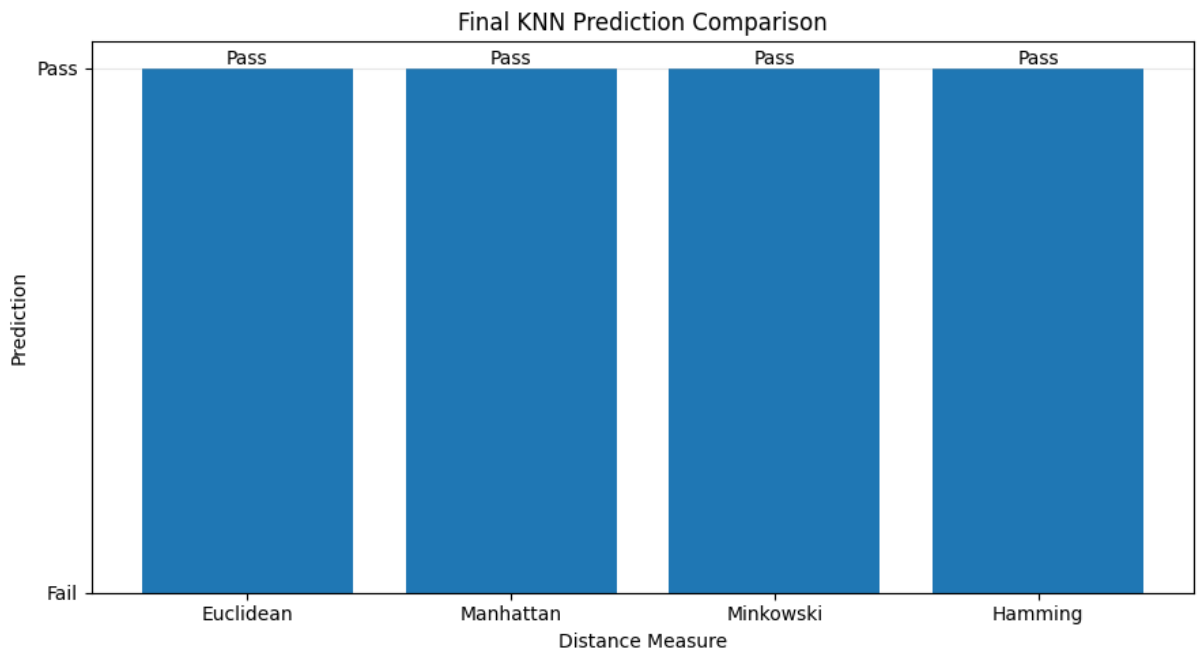
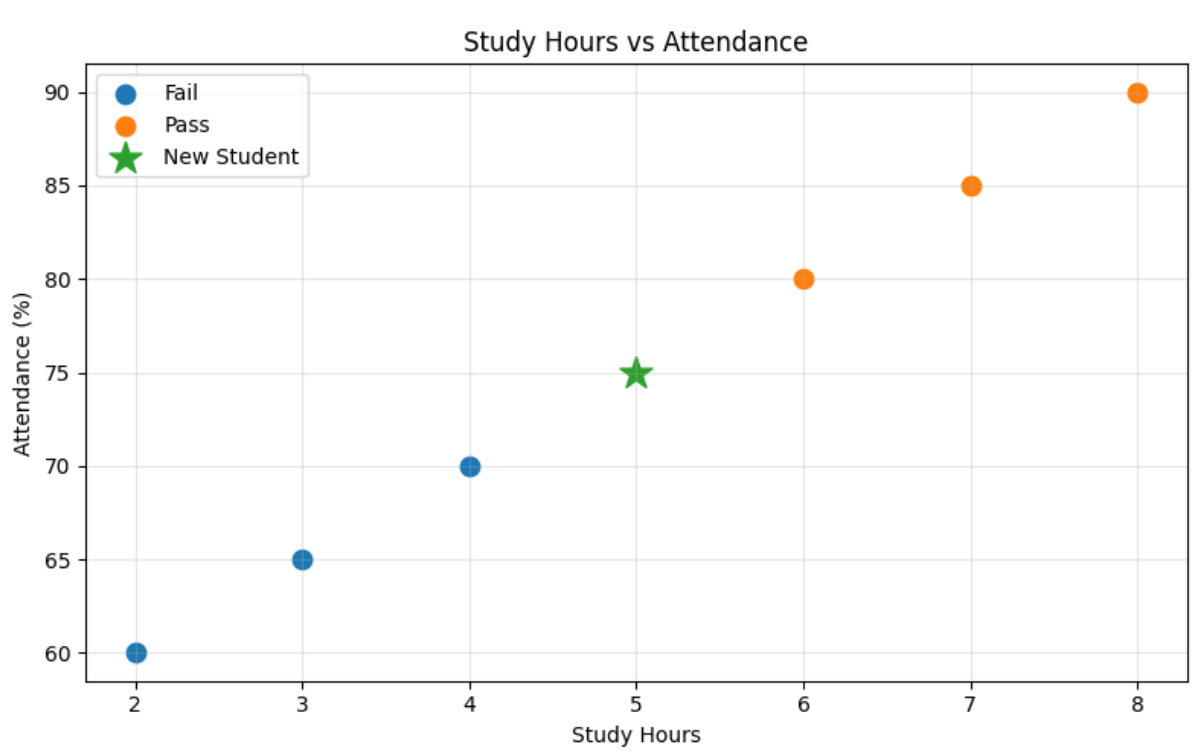
FINAL PREDICTIONS

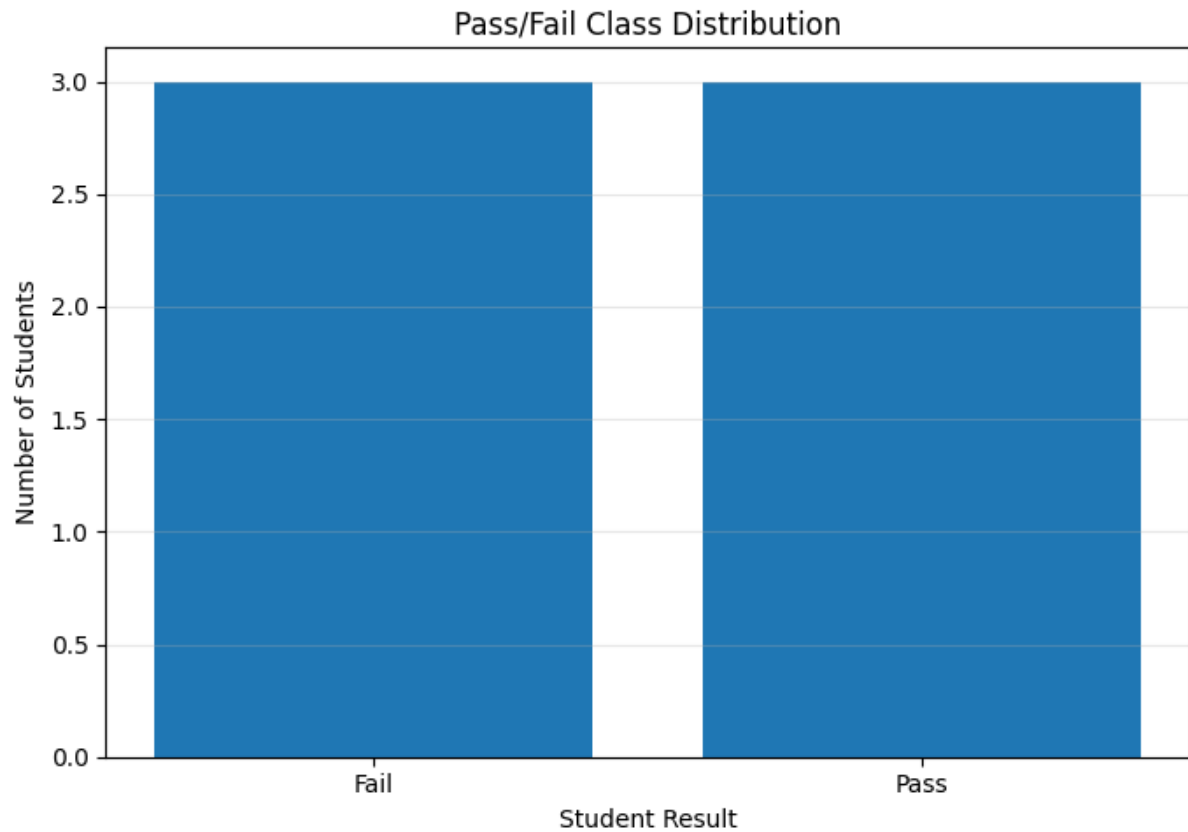
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Euclidean	: Pass
Manhattan	: Pass
Minkowski	: Pass
Hamming	: Pass









Question 2: Online Customer Purchase Prediction

An e-commerce company wants to predict whether a customer will Buy or Not Buy a product.

Customer	Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer	Class
C1	10	3	0	0	0	Not Buy
C2	15	4	1	0	0	Not Buy
C3	25	7	1	1	1	Buy
C4	30	8	1	1	1	Buy
C5	18	5	0	1	0	Not Buy
C6	28	9	1	1	1	Buy

A new customer has:

Browsing Time	Items Viewed	Coupon Used	Added to Cart	Returning Customer
22	6	1	1	0

Using KNN with $K = 3$, classify the new customer as Buy or Not Buy using all four distance measures.

⑤ Question 8: Online Customer Purchase Prediction

New Customer $X = (22, 6, 1, 1, 0)$, $k=3$

Euclidean :-

$C_3 \rightarrow \text{Buy}$, $C_5 \rightarrow \text{not Buy}$, $C_6 \rightarrow \text{Buy}$

Formula :-

$$D_E = \sqrt{\sum (x_i - y_i)^2}$$

$$C_1 = \sqrt{(22-10)^2 + (6-3)^2 + (1-0)^2 + (1-0)^2 + (0-0)^2} = \sqrt{155} = 12.4499$$

$$C_2 = \sqrt{59} = 7.685, \quad C_3 = \sqrt{11} = 3.3166$$

$$C_4 = \sqrt{69} = 8.3016, \quad C_5 = \sqrt{16} = 4.2426, \quad C_6 = \sqrt{46} = 6.7823$$

Voting :- Buy = 2, Not Buy = 1

Prediction = Buy

Manhattan Distance

Formula :- $D_M = \sum |x_i - y_i|$

Customer	Calculation	Distance
C_1	$12+7+1+1+0$	17
C_2	$7+2+0+1+0$	10
C_3	$3+1+0+0+1$	5
C_4	$8+2+0+0+1$	11
C_5	$4+1+1+0+1$	6
C_6	$6+3+0+0+1$	10

3-Nearest :- C_3, C_5, C_2

C_2 and C_6 have same distance, so using original order C_2 is selected

Value)

Buy = 1, Not Buy = 2

Prediction = Not Buy

Minkowski Distance

Given $p=3$

Formula :-

$$D_M = (\sum |x_i - y_i|^p)^{1/p}$$

Calculations

Customer

Calculation

Distance

C_1	$(10^3 + 3^3 + 1^3 + 1^3)^{1/3}$	12.0668
C_2	$(7^3 + 2^3 + 1^3)^{1/3}$	7.0607
C_3	$(3^3 + 1^3 + 1^3)^{1/3}$	3.0923
C_4	$(8^3 + 2^3 + 1^3)^{1/3}$	8.0466
C_5	$(4^3 + 1^3 + 1^3)^{1/3}$	4.0412
C_6	$(6^3 + 3^3 + 1^3)^{1/3}$	6.2488

3 Nearest $\rightarrow C_3, C_5, C_6$

Voting :- Buy = 2, Not Buy = 1

Prediction = Buy

Hamming Distance

(Coupon, Cost, Returning)

$X = (1, 1, 0)$

Formula :- $D_H = \text{No. of different binary values}$

<u>Customer</u>	<u>Binary Value</u>	<u>Hamming</u>
C_1	(0, 0, 0)	2
C_2	(1, 0, 0)	1
C_3	(1, 1, 1)	1
C_4	(1, 1, 1)	1
C_5	(0, 1, 0)	1
C_6	(1, 1, 1)	1

using original order: C_2, C_3, C_4

class: Not Buy, Buy, Buy

$\therefore$ Prediction = Buy

Output:

ONLINE CUSTOMER PURCHASE DATA

Customer Used_Coupon	Age \	Previous_Purchases	Mobile_User	Added_to_Cart	
0	C1	22	2	1	0
1					
1	C2	25	3	1	1
1					
2	C3	35	5	1	1
0					
3	C4	40	6	1	1
0					
4	C5	28	4	0	1
1					
5	C6	45	7	1	1
0					

	Class
0	No Purchase
1	No Purchase
2	Purchase
3	Purchase
4	No Purchase
5	Purchase

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DISTANCE CALCULATIONS

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Customer	Class	Euclidean	Manhattan	Minkowski	Hamming
C1	No Purchase	8.3666	12	8.0517	2
C2	No Purchase	5.1962	7	5.0265	1
C3	Purchase	5.0990	6	5.0133	0
C4	Purchase	10.1980	12	10.0266	0
C5	No Purchase	2.4495	4	2.1544	2
C6	Purchase	15.2971	18	15.0399	0

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EUCLIDEAN DISTANCE

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3 Nearest Neighbours:

Customer	Class	Distance
C5	No Purchase	2.449490
C3	Purchase	5.099020
C2	No Purchase	5.196152

Majority Voting:

Class

No Purchase 2

Purchase 1

Name: count, dtype: int64

Prediction: No Purchase

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MANHATTAN DISTANCE

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3 Nearest Neighbours:

Customer	Class	Distance
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C5 No Purchase	4
C3 Purchase	6
C2 No Purchase	7

Majority Voting:

Class

No Purchase 2

Purchase 1

Name: count, dtype: int64

Prediction: No Purchase

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MINKOWSKI DISTANCE (p = 3)
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3 Nearest Neighbours:

Customer	Class	Distance
C5 No Purchase		2.154435
C3 Purchase		5.013298
C2 No Purchase		5.026526

Majority Voting:

Class

No Purchase 2

Purchase 1

Name: count, dtype: int64

Prediction: No Purchase

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HAMMING DISTANCE
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3 Nearest Neighbours:

Customer	Class	Distance
C4 Purchase		0
C3 Purchase		0
C6 Purchase		0

Majority Voting:

Class

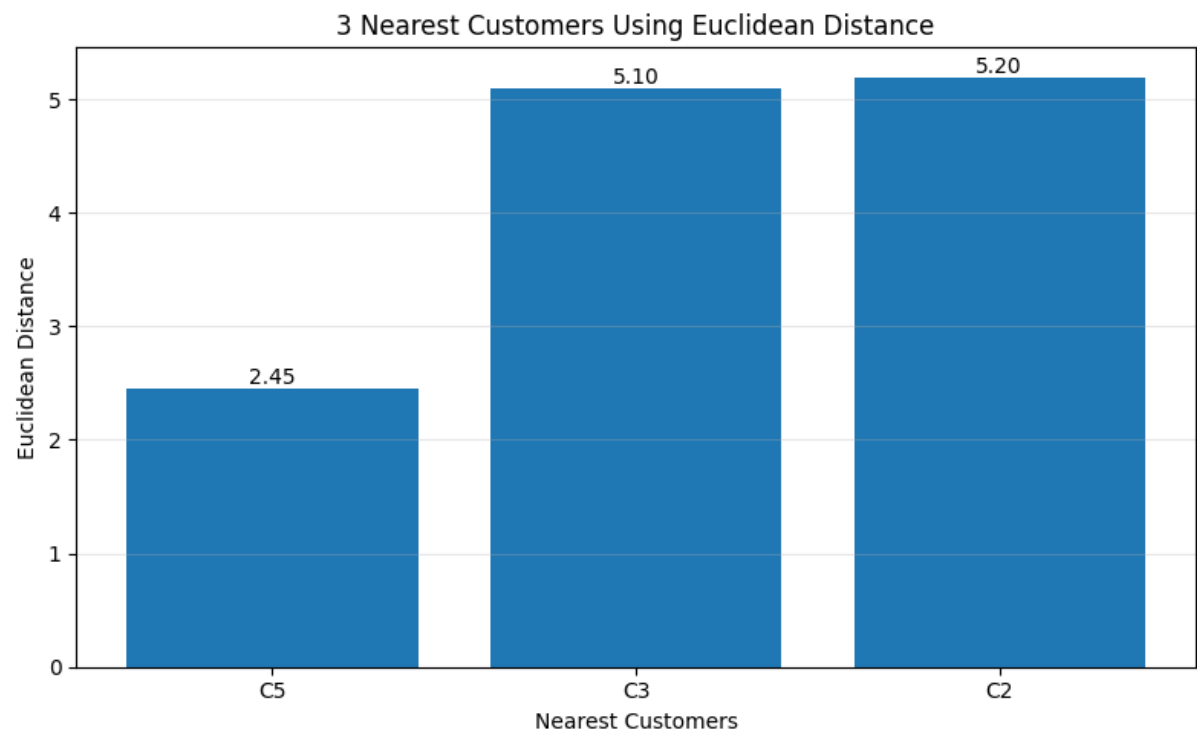
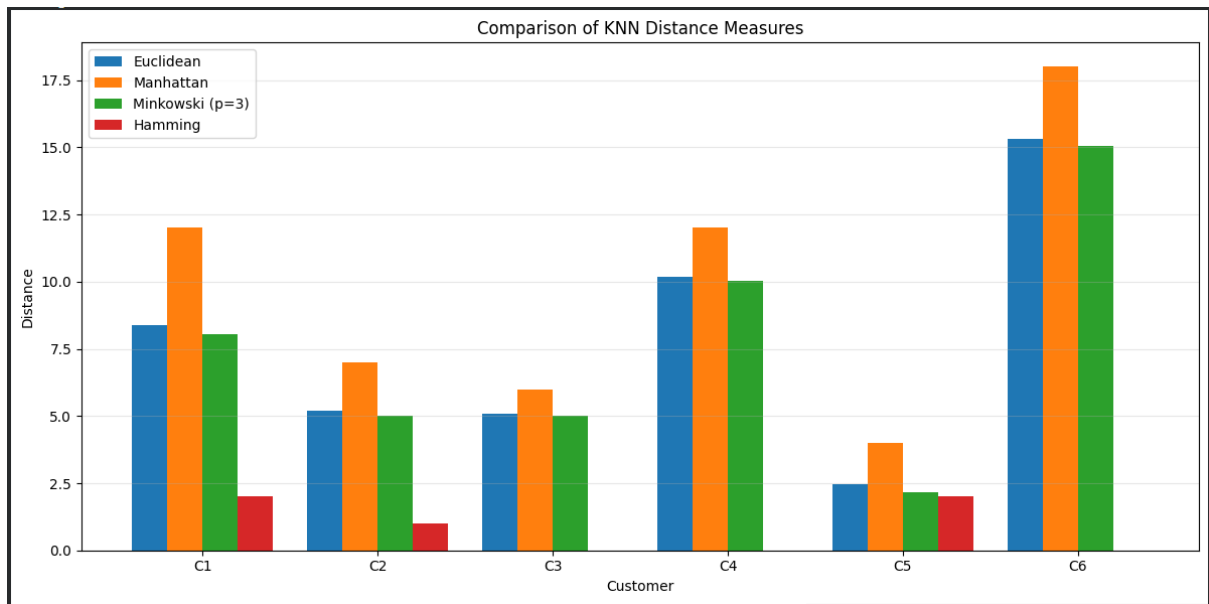
Purchase 3

Name: count, dtype: int64

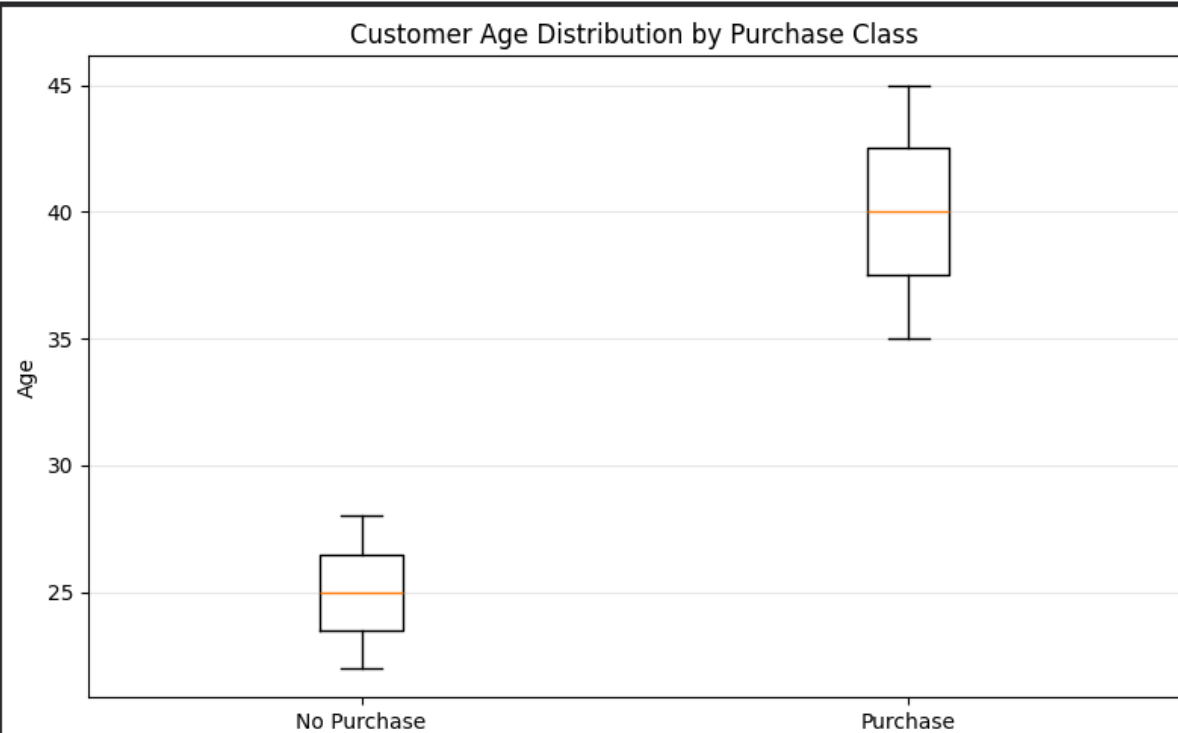
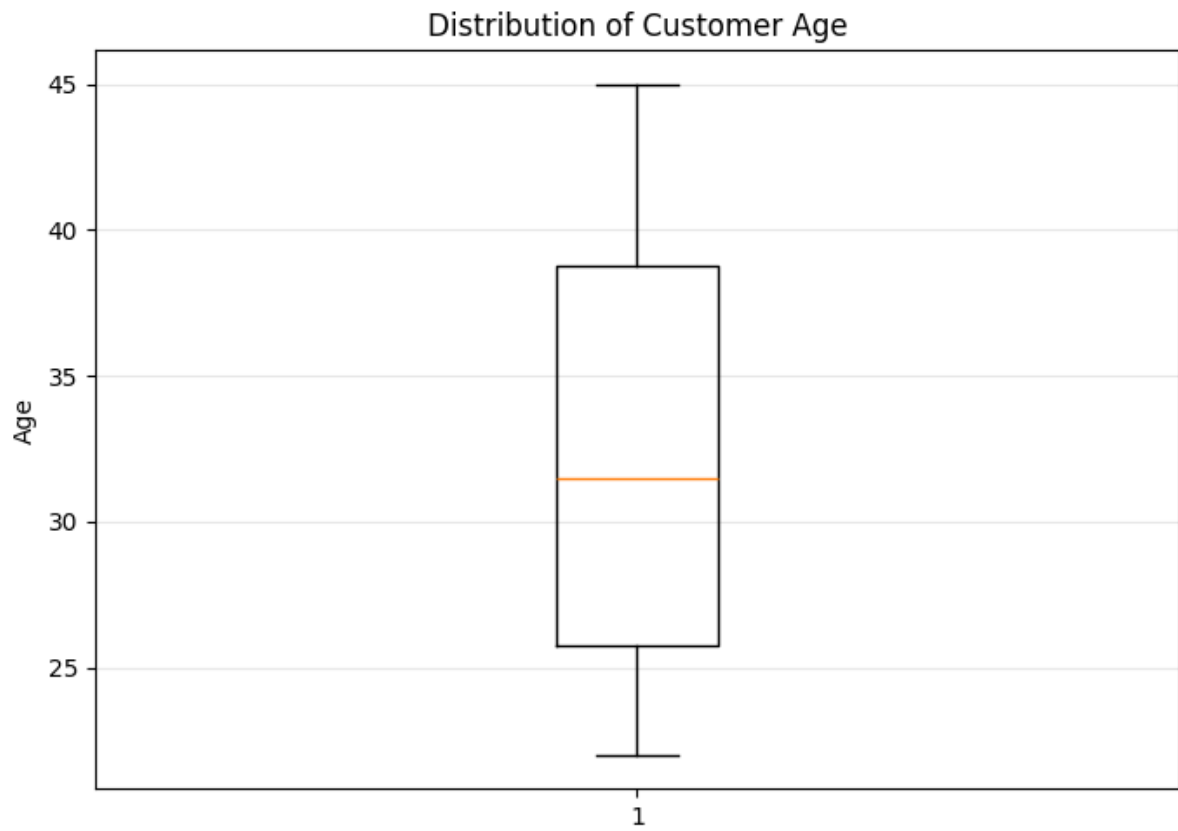
Prediction: Purchase

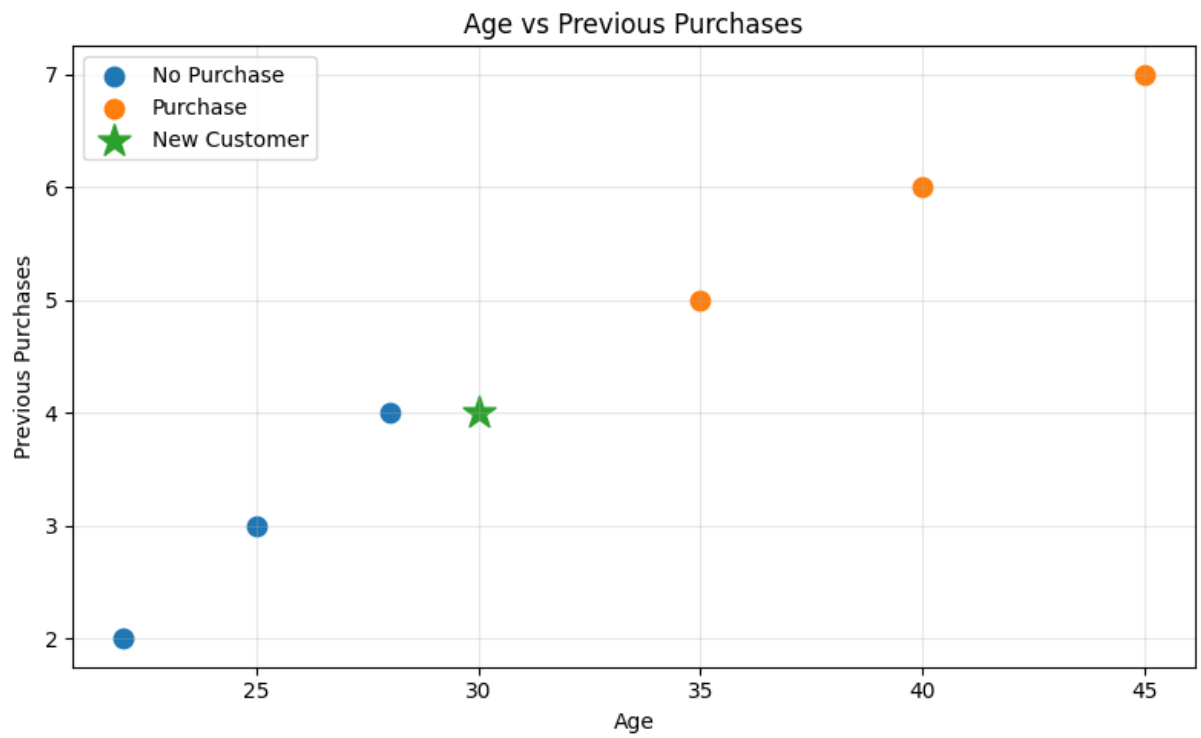
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FINAL PREDICTIONS
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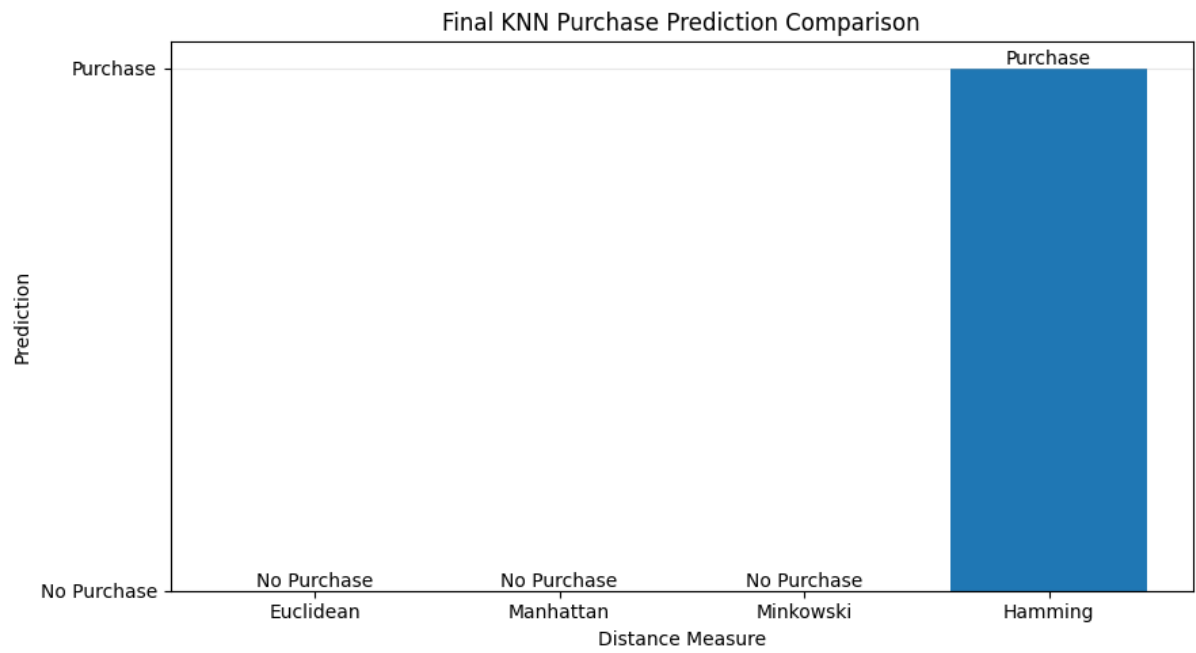
Euclidean	: No Purchase
Manhattan	: No Purchase
Minkowski	: No Purchase
Hamming	: Purchase



Nearest customers







Question 3: Medical Risk Classification

A hospital wants to classify whether a patient has High Risk or Low Risk based on medical measurements and symptoms.

Patient	Age	Blood Sugar Level	Fever	Cough	Fatigue	Class
P1	25	90	0	0	0	Low Risk
P2	30	95	0	1	0	Low Risk
P3	50	160	1	1	1	High Risk
P4	55	170	1	1	1	High Risk
P5	40	120	0	1	1	Low Risk
P6	60	180	1	1	1	High Risk

A new patient has:

Age	Blood Sugar Level	Fever	Cough	Fatigue
45	150	1	1	0

Using KNN with $K = 3$, classify the patient as High Risk or Low Risk using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 2: Medical Risk classification :-

$X = (45, 150, 1, 1, 0)$

where $k=3$

Euclidean Distance :-

Formula :- $D_E = \sqrt{\sum (x_i - y_i)^2}$

Patient	Calculation	Distance
P_1	$\sqrt{40^2 + 60^2 + 1^2 + 1^2 + 0^2}$	63.2535
P_2	$\sqrt{11^2 + 55^2 + 1^2 + 0^2 + 0^2}$	57.0175
P_3	$\sqrt{5^2 + 10^2 + 0^2 + 0^2 + 1^2}$	11.2250
P_4	$\sqrt{10^2 + 20^2 + 0^2 + 0^2 + 1^2}$	22.3607
P_5	$\sqrt{5^2 + 30^2 + 1^2 + 0^2 + 1^2}$	30.8622
P_6	$\sqrt{15^2 + 30^2 + 0^2 + 0^2 + 1^2}$	33.5710

5 Nearest

class: $P_1 \rightarrow$ High Risk, $P_4 \rightarrow$ High Risk, $P_5 \rightarrow$ Low Risk

High Risk = 2, Low Risk = 1

Prediction = High Risk

Manhattan Distance

Formula: $D_M = \sum |x_i - y_i|$

Patient	Calculation	Distance
P_1	$20 + 60 + 1 + 1 + 0$	82
P_2	$15 + 55 + 1 + 0 + 0$	71
P_3	$5 + 10 + 0 + 0 + 1$	16
P_4	$10 + 20 + 0 + 0 + 1$	31
P_5	$5 + 30 + 1 + 0 + 1$	37
P_6	$15 + 30 + 0 + 0 + 1$	46

3 Nearest Neighbours: P_3, P_4, P_5

Voting: High Risk = 2, Low Risk = 1

Prediction = High Risk

Minkowski Distance ($p=3$)

Formula: $D_M = (\sum |x_i - y_i|^3)^{1/3}$

Patient	Calculation	Distance
P_1	$(20^3 + 60^3 + 1^3 + 1^3)^{1/3}$	60.9598
P_2	$(15^3 + 55^3 + 1^3)^{1/3}$	55.5444
P_3	$(5^3 + 10^3 + 1^3)^{1/3}$	10.7877
P_4	$(10^3 + 20^3 + 1^3)^{1/3}$	20.8091
P_5	$(5^3 + 30^3 + 1^3 + 1^3)^{1/3}$	30.0926
P_6	$(15^3 + 30^3 + 1^3)^{1/3}$	30.1376

3. Nearest Neighbours

P_3, P_4, P_5

Classes: High Risk, High Risk, Low Risk

Prediction = High Risk

Hamming Distance :-

(Fever, Cough, Fatigue)

$$X_B = (1, 1, 0)$$

Formula :-

D_H = No. of different binary values

Patient	Binary values	Hamming
P_1	(0, 0, 0)	2
P_2	(0, 1, 0)	1
P_3	(1, 1, 1)	1
P_4	(1, 1, 1)	1
P_5	(0, 1, 1)	2
P_6	(1, 1, 1)	1

There is a tie at between P_2, P_3, P_4 and P_6

original order : P_2, P_3, P_4

Classes :-

$P_2 \rightarrow$ Low Risk, $P_3 \rightarrow$ High Risk, $P_4 \rightarrow$ High Risk

$\therefore$ High Risk = 2, Low Risk = 1

Prediction = High Risk

Output:

MEDICAL RISK DATA

	Patient	Age	Blood_Sugar_Level	Fever	Cough	Fatigue	Class
0	P1	25	90	0	0	0	Low Risk
1	P2	30	95	0	1	0	Low Risk
2	P3	50	160	1	1	1	High Risk
3	P4	55	170	1	1	1	High Risk
4	P5	40	120	0	1	1	Low Risk
5	P6	60	180	1	1	1	High Risk

DISTANCE CALCULATIONS

Patient	Class	Euclidean	Manhattan	Minkowski	Hamming
P1	Low Risk	63.2614	82	60.7320	2
P2	Low Risk	57.0175	71	55.3695	1
P3	High Risk	11.2250	16	10.4035	1
P4	High Risk	22.3830	31	20.8016	1
P5	Low Risk	30.4467	37	30.0470	2
P6	High Risk	33.5559	46	31.2016	1

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Patient	Class	Distance
P3	High Risk	11.224972
P4	High Risk	22.383029
P5	Low Risk	30.446675

Majority Voting:

Class

High Risk 2

Low Risk 1

Name: count, dtype: int64

Prediction: High Risk

MANHATTAN DISTANCE

3 Nearest Neighbours:

Patient	Class	Distance
P3	High Risk	16
P4	High Risk	31
P5	Low Risk	37

Majority Voting:

Class

High Risk 2

Low Risk 1

Name: count, dtype: int64

Prediction: High Risk

MINKOWSKI DISTANCE (p = 3)

3 Nearest Neighbours:
Patient Class Distance
 P3 High Risk 10.403500
 P4 High Risk 20.801609
 P5 Low Risk 30.046963

Majority Voting:
Class
High Risk 2
Low Risk 1
Name: count, dtype: int64

Prediction: High Risk

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HAMMING DISTANCE
=====

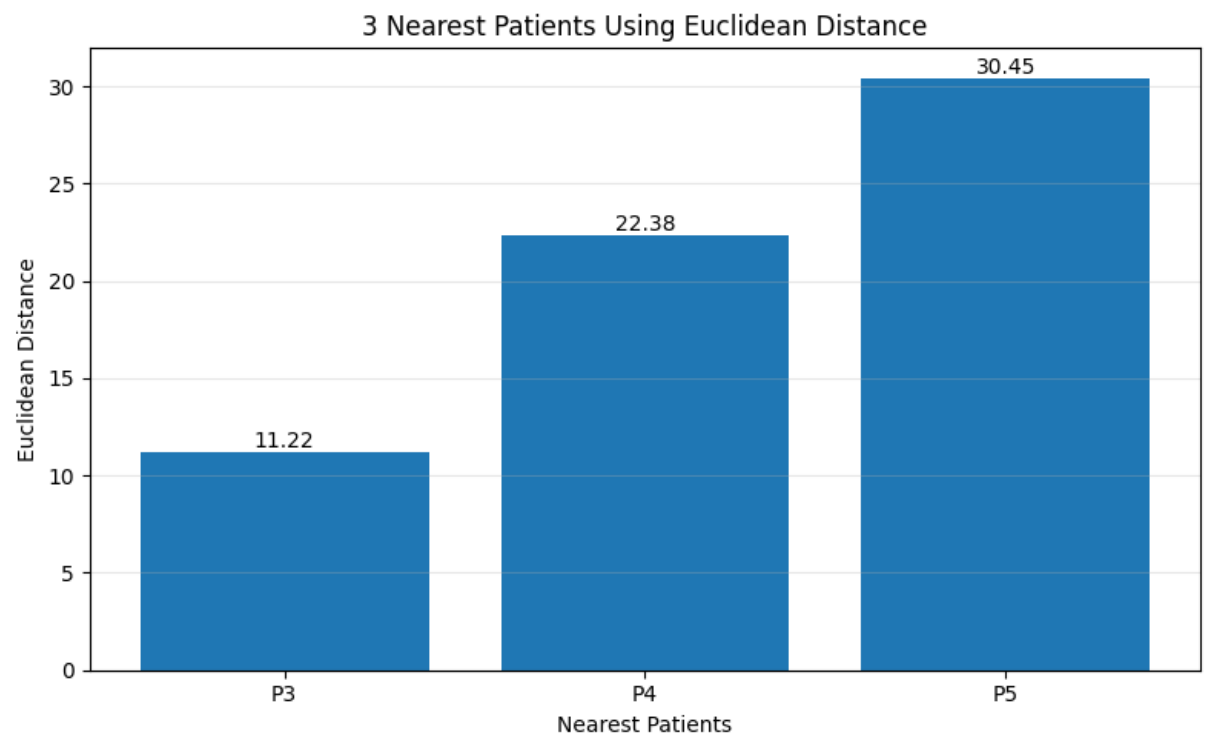
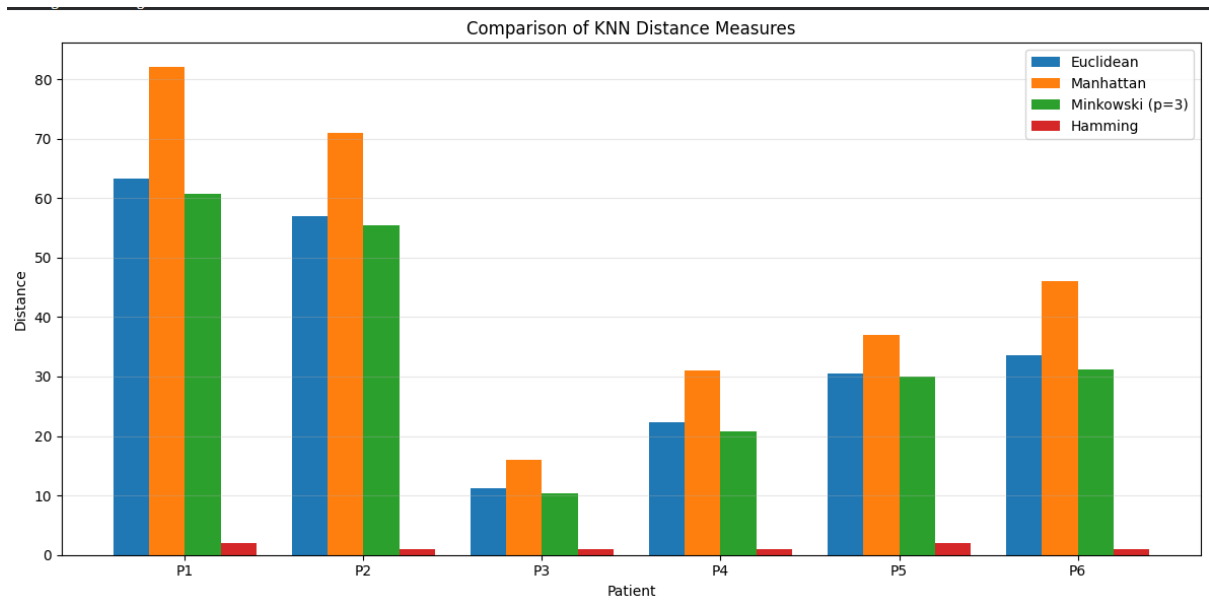
3 Nearest Neighbours:
Patient Class Distance
 P2 Low Risk 1
 P3 High Risk 1
 P4 High Risk 1

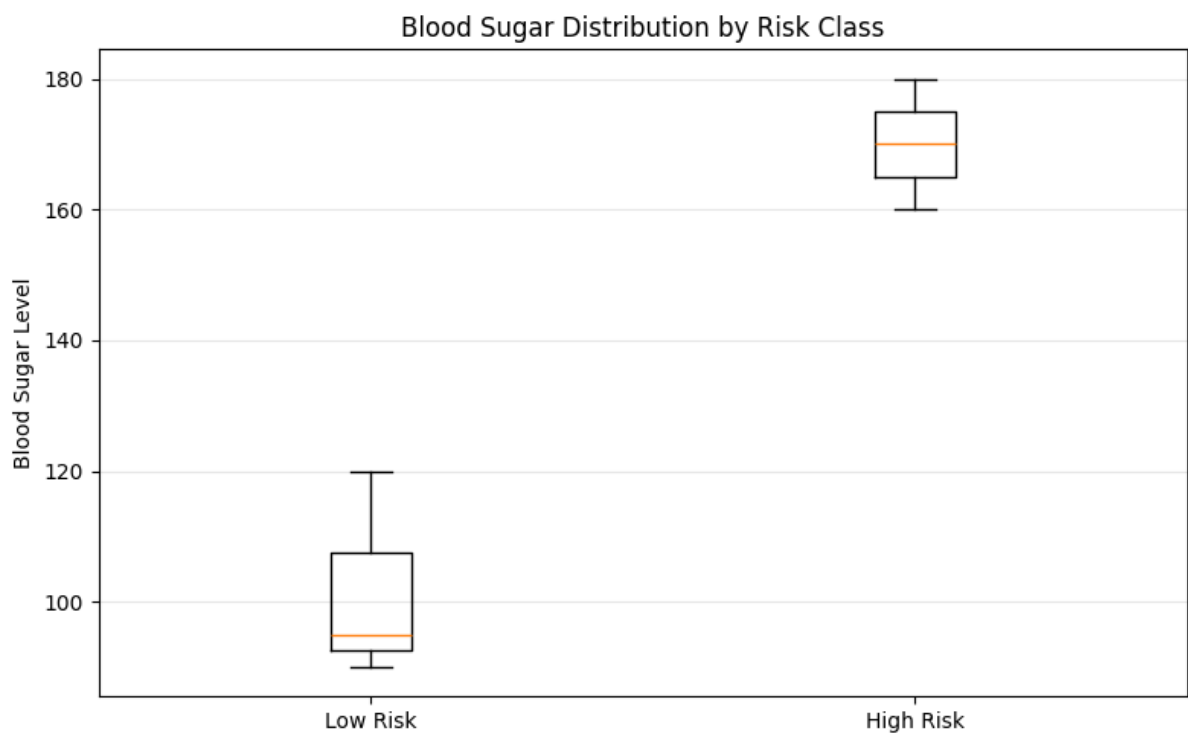
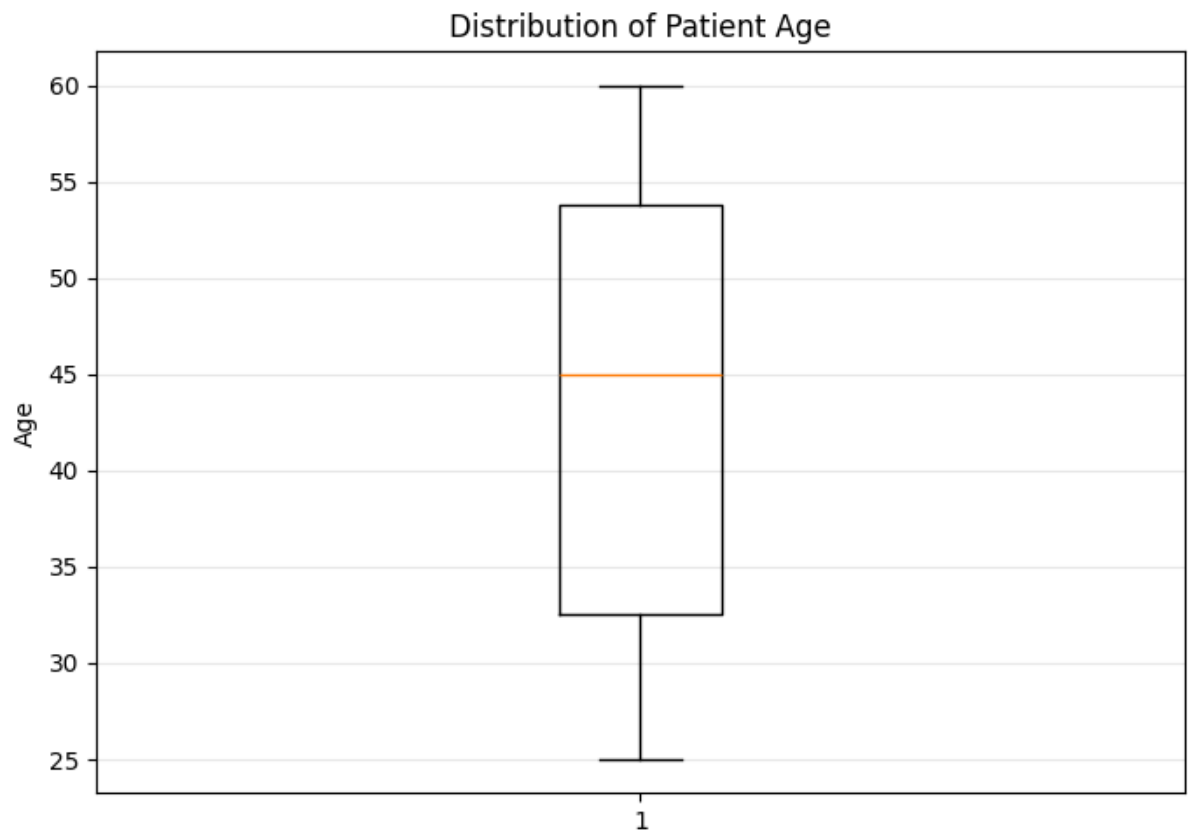
Majority Voting:
Class
High Risk 2
Low Risk 1
Name: count, dtype: int64

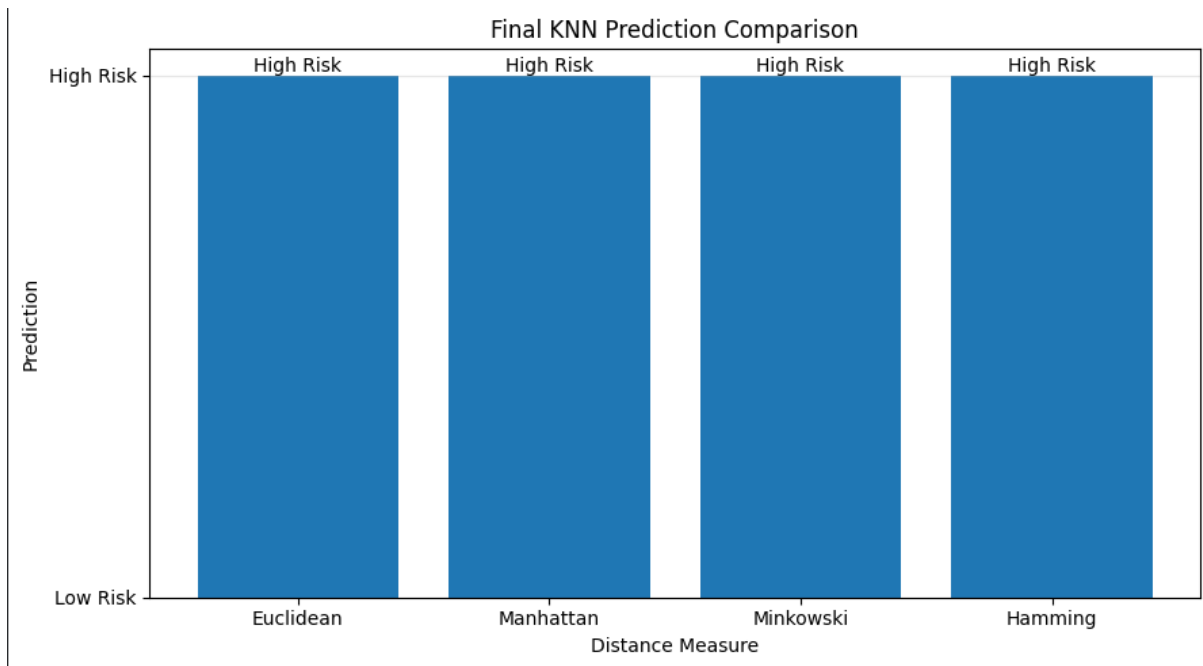
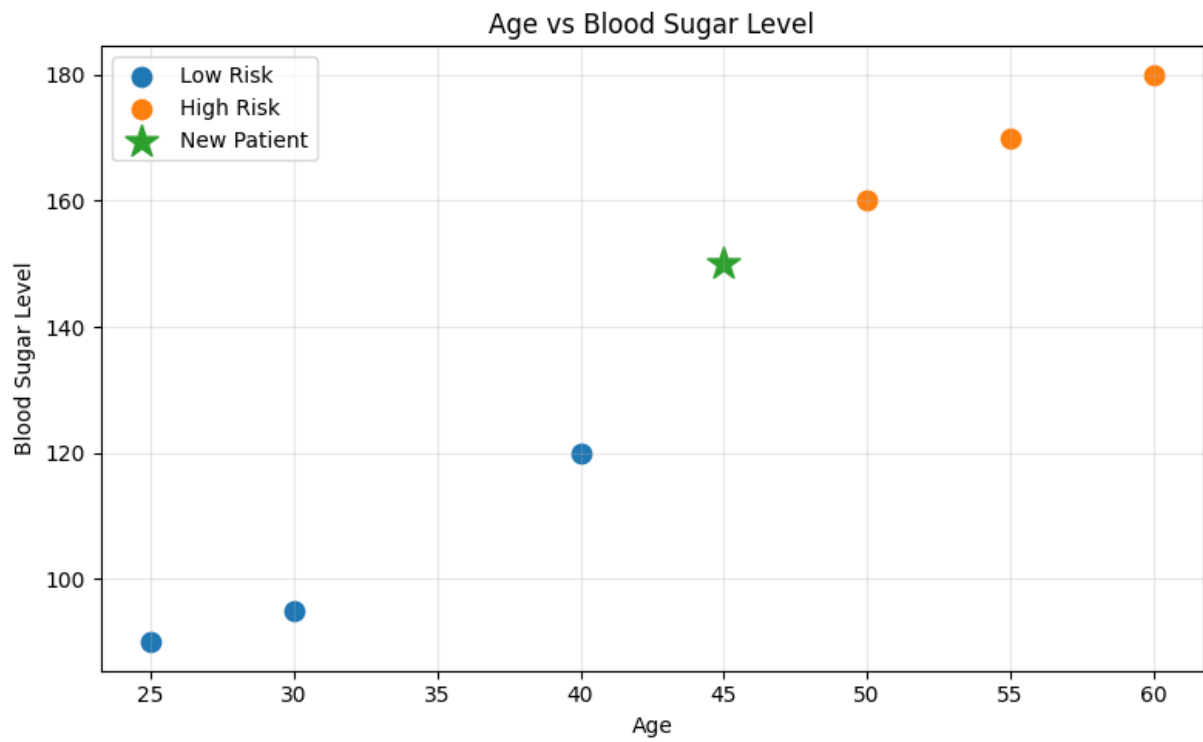
Prediction: High Risk

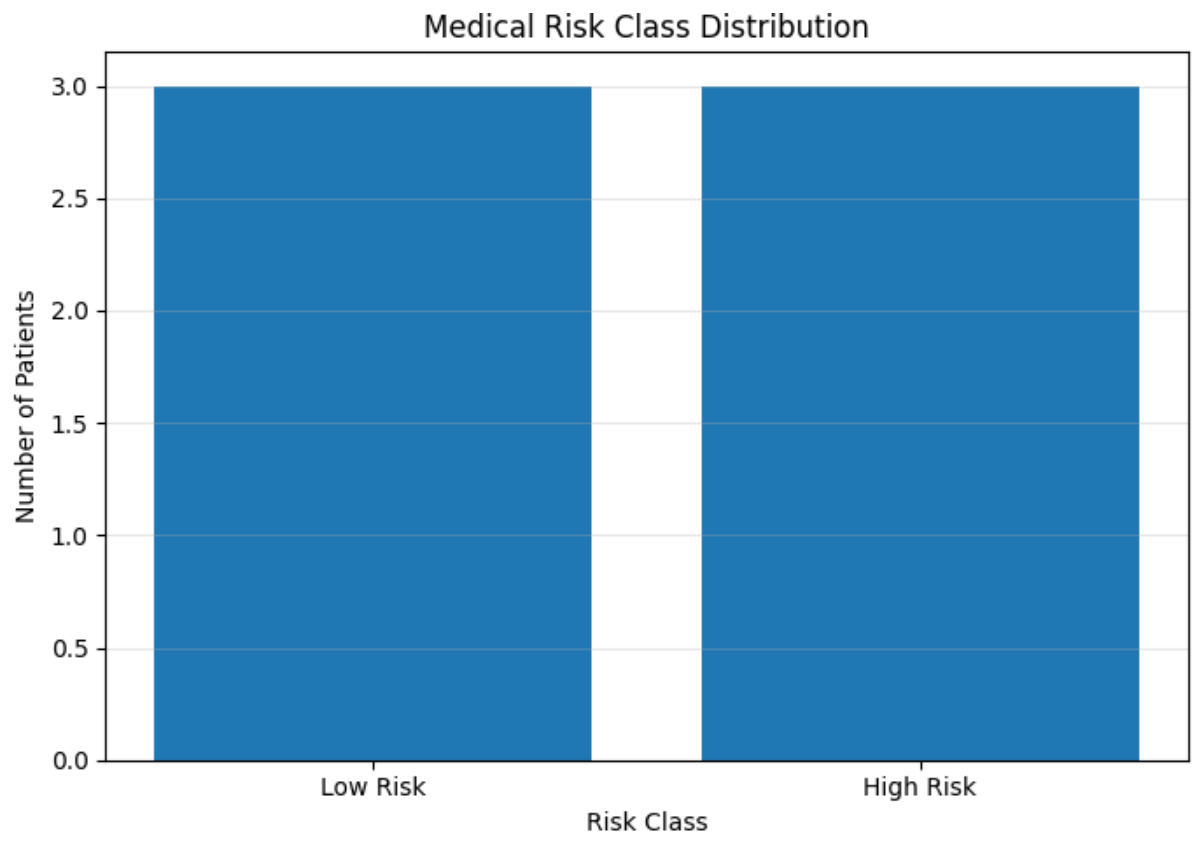
=====
FINAL PREDICTIONS
=====

Euclidean : High Risk
Manhattan : High Risk
Minkowski : High Risk
Hamming : High Risk









Question 4: Fruit Classification

A fruit shop wants to classify a fruit as Apple or Orange based on physical and binary characteristics.

Fruit	Weight	Diameter	Red Color	Citrus Smell	Smooth Skin	Class
F1	150	7	1	0	1	Apple
F2	160	8	1	0	1	Apple
F3	180	9	0	1	0	Orange
F4	200	10	0	1	0	Orange
F5	155	7.5	1	0	1	Apple
F6	190	9.5	0	1	0	Orange

A new fruit has:

Weight	Diameter	Red Color	Citrus Smell	Smooth Skin
170	8.5	1	0	1

Using KNN with $K = 3$, classify the fruit as Apple or Orange using all four distance measures.

Question 4: 19/01/2021

Given New Fruit

$$X = (170, 8.5, 1, 0, 1)$$

where, $k=3$

1. Euclidean Distance

Formula :-

$$D_e = \sqrt{\sum (x_i - y_i)^2}$$

<u>Fruit</u>	<u>Calculation</u>	<u>Distance</u>
F_1	$\sqrt{20^2 + 15^2 + 0^2 + 0^2 + 0^2}$	20.0562
F_2	$\sqrt{10^2 + 0.5^2 + 0^2 + 0^2 + 0^2}$	10.0125
F_3	$\sqrt{10^2 + 0.5^2 + 1^2 + 1^2 + 1^2}$	10.1612
F_4	$\sqrt{30^2 + 1.5^2 + 1^2 + 1^2 + 1^2}$	30.0674
F_5	$\sqrt{15^2 + 1^2 + 0^2 + 0^2 + 0^2}$	15.0323
F_6	$\sqrt{20^2 + 1^2 + 1^2 + 1^2 + 1^2}$	20.0998

3 Nearest Neighbours : F_2, F_3, F_5

Classes: $F_2 \rightarrow$ Apple, $F_3 \rightarrow$ Orange, $F_5 \rightarrow$ Apple

Apple = 2, Orange = 1

Prediction = Apple

2. Manhattan Distance

Formula :-

$$D_m = \sum |x_i - y_i|$$

$$F_1 = 20 + 1.5 + 0 + 0 + 0 = 21.5$$

$$F_2 = 10 + 0.5 + 0 + 0 + 0 = 10.5$$

$$F_3 = 10 + 0.5 + 1 + 1 + 1 = 13.5$$

$$F_4 = 30 + 1.5 + 1 + 1 + 1 = 34.5$$

$$F_5 = 15 + 1 + 0 + 0 + 0 = 16$$

$$F_6 = 20 + 1 + 1 + 1 + 1 = 24$$

3 Nearest Neighbours: F_1, F_3, F_5

Voting:-

Apple = 2, Orange = 1

Prediction = Apple

3. Minkowski Distance ($p=3$)

Formula:-

$$D_M = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

Fruit

$$F_1 = 20.0018$$

$$F_2 = 10.0004$$

$$F_3 = 10.0104$$

$$F_4 = 30.0029$$

$$F_5 = 15.0015$$

$$F_6 = 20.0033$$

3 Nearest Neighbours

F_2, F_3, F_5

Prediction = Apple

4. Hamming Distance

New fruit (1,0,1)

$$F_1 \rightarrow 0, F_2 \rightarrow 0, F_3 \rightarrow 3, F_4 \rightarrow 3, F_5 \rightarrow 0, F_6 \rightarrow 3$$

Prediction = Apple

Output:

FRUIT DATA

	Fruit	Weight	Diameter	Red_Color	Citrus_Smell	Smooth_Skin	Class
0	F1	150	7.0	1	0	1	Apple
1	F2	160	8.0	1	0	1	Apple
2	F3	180	9.0	0	1	0	Orange
3	F4	200	10.0	0	1	0	Orange
4	F5	155	7.5	1	0	1	Apple
5	F6	190	9.5	0	1	0	Orange

DISTANCE CALCULATIONS

Fruit	Class	Euclidean	Manhattan	Minkowski	Hamming
F1	Apple	20.0562	21.5	20.0028	0
F2	Apple	10.0125	10.5	10.0004	0
F3	Orange	10.1612	13.5	10.0104	3
F4	Orange	30.0874	34.5	30.0024	3
F5	Apple	15.0333	16.0	15.0015	0
F6	Orange	20.0998	24.0	20.0033	3

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Fruit	Class	Distance
F2	Apple	10.012492
F3	Orange	10.161201
F5	Apple	15.033296

Majority Voting:

Class

Apple 2

Orange 1

Name: count, dtype: int64

Prediction: Apple

MANHATTAN DISTANCE

3 Nearest Neighbours:

Fruit	Class	Distance
F2	Apple	10.5
F3	Orange	13.5
F5	Apple	16.0

Majority Voting:

Class

Apple 2

Orange 1

Name: count, dtype: int64

Prediction: Apple

MINKOWSKI DISTANCE (p = 3)

3 Nearest Neighbours:

Fruit	Class	Distance
F2	Apple	10.000417
F3	Orange	10.010406
F5	Apple	15.001481

Majority Voting:

Class

Apple 2

Orange 1

Name: count, dtype: int64

Prediction: Apple

=====

HAMMING DISTANCE

=====

3 Nearest Neighbours:

Fruit	Class	Distance
F1	Apple	0
F2	Apple	0
F5	Apple	0

Majority Voting:

Class

Apple 3

Name: count, dtype: int64

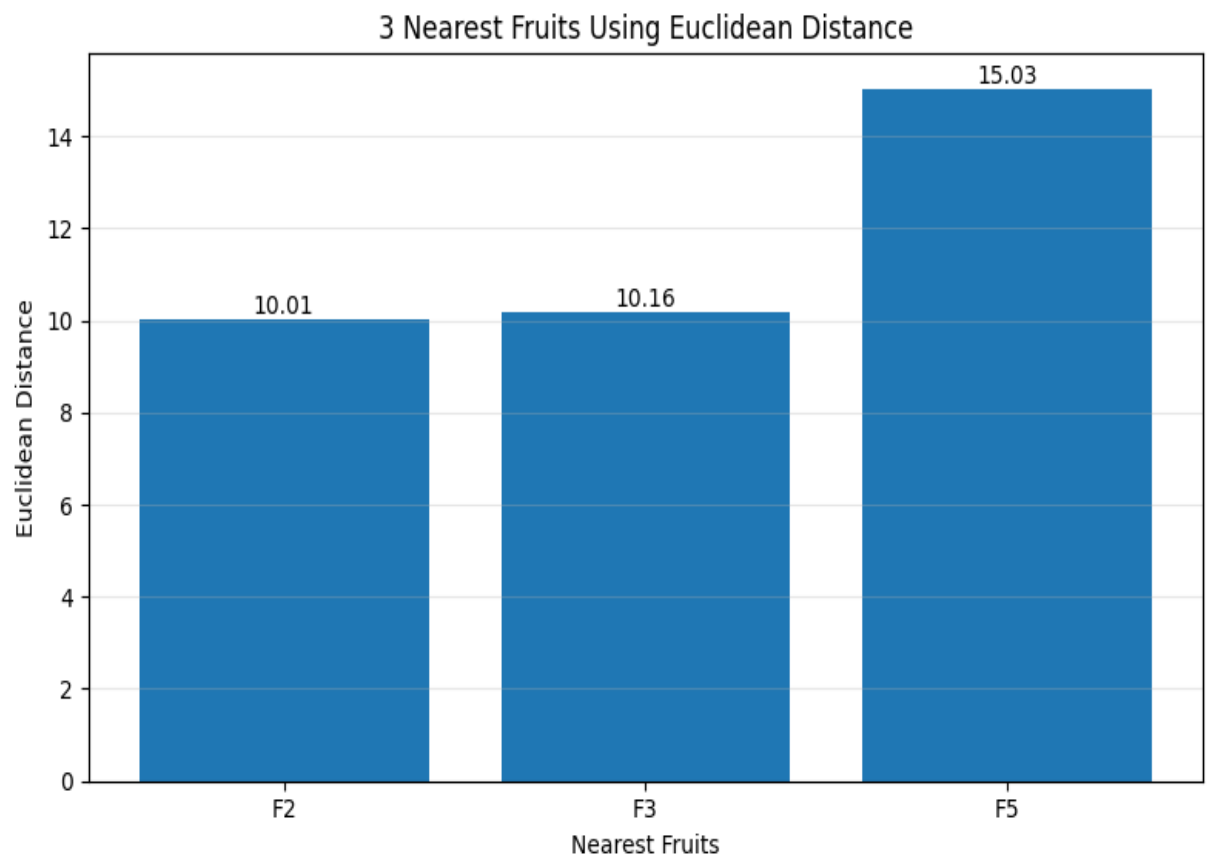
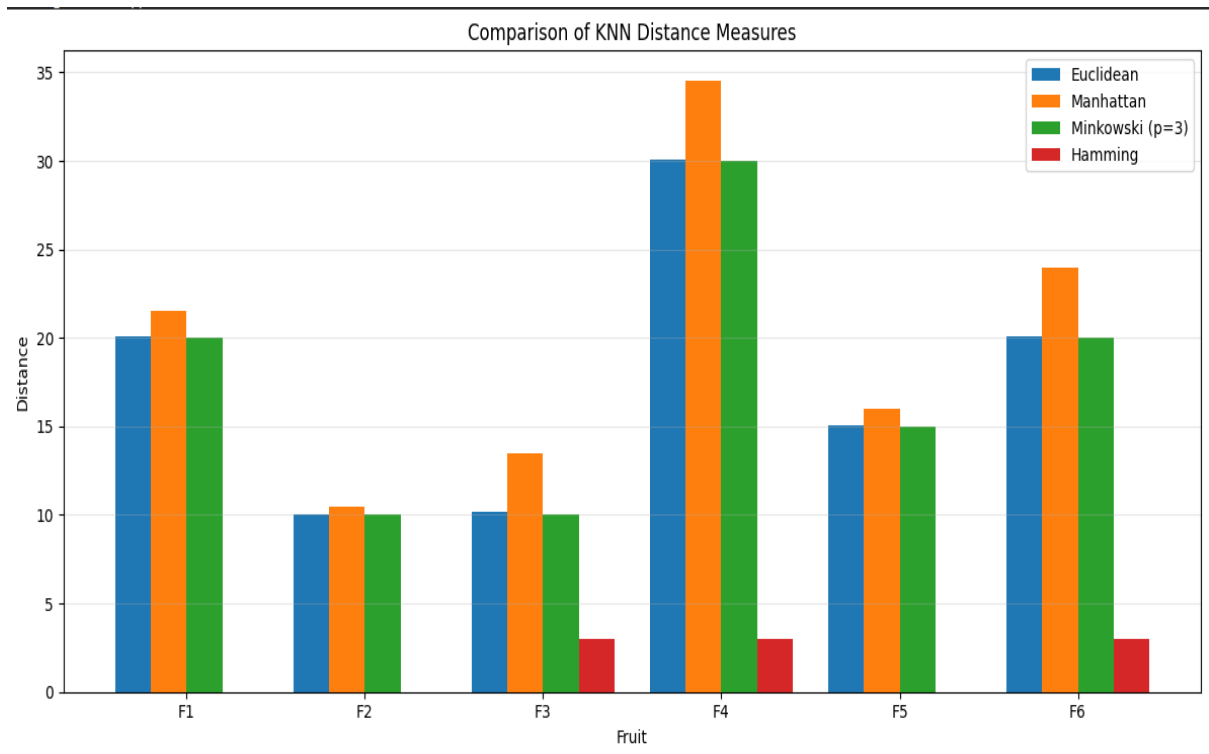
Prediction: Apple

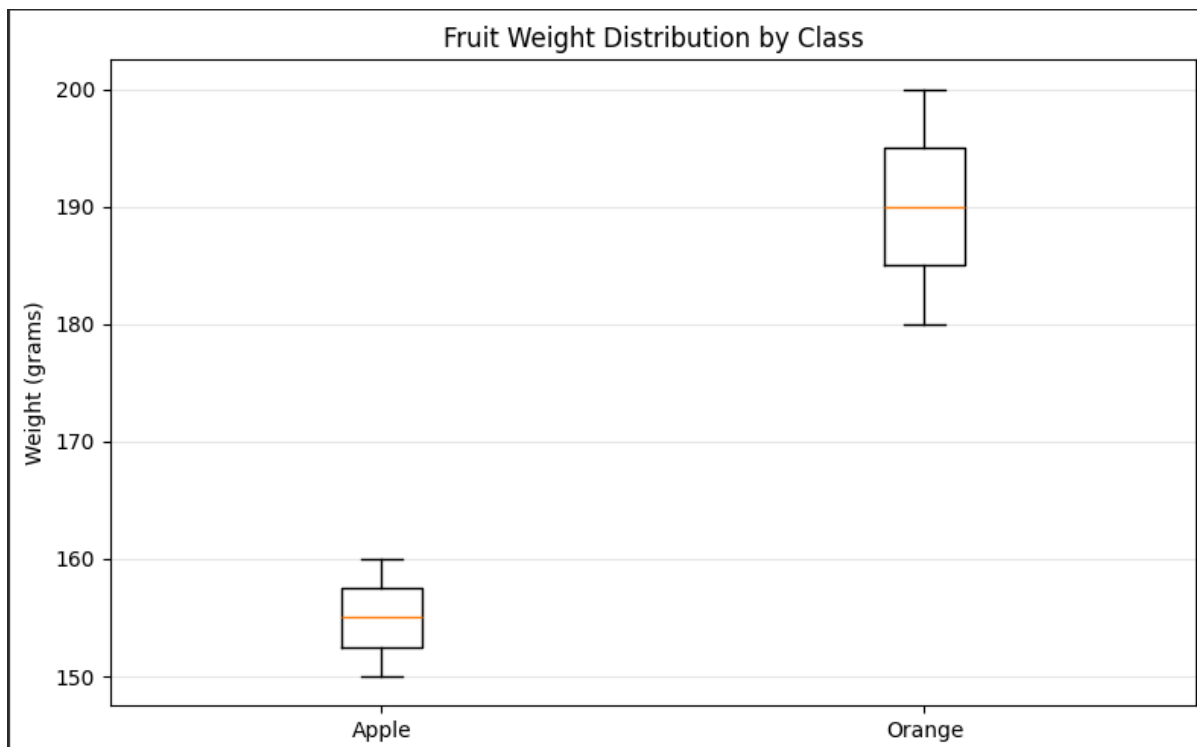
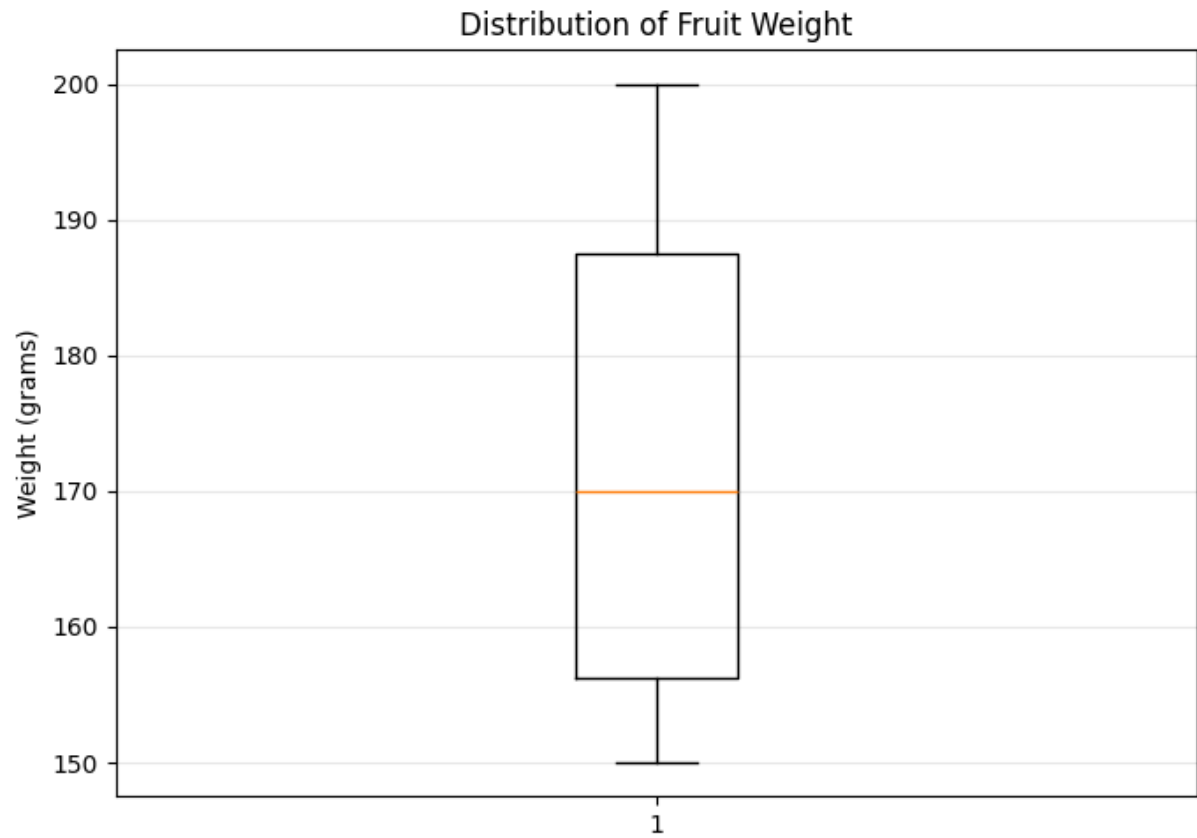
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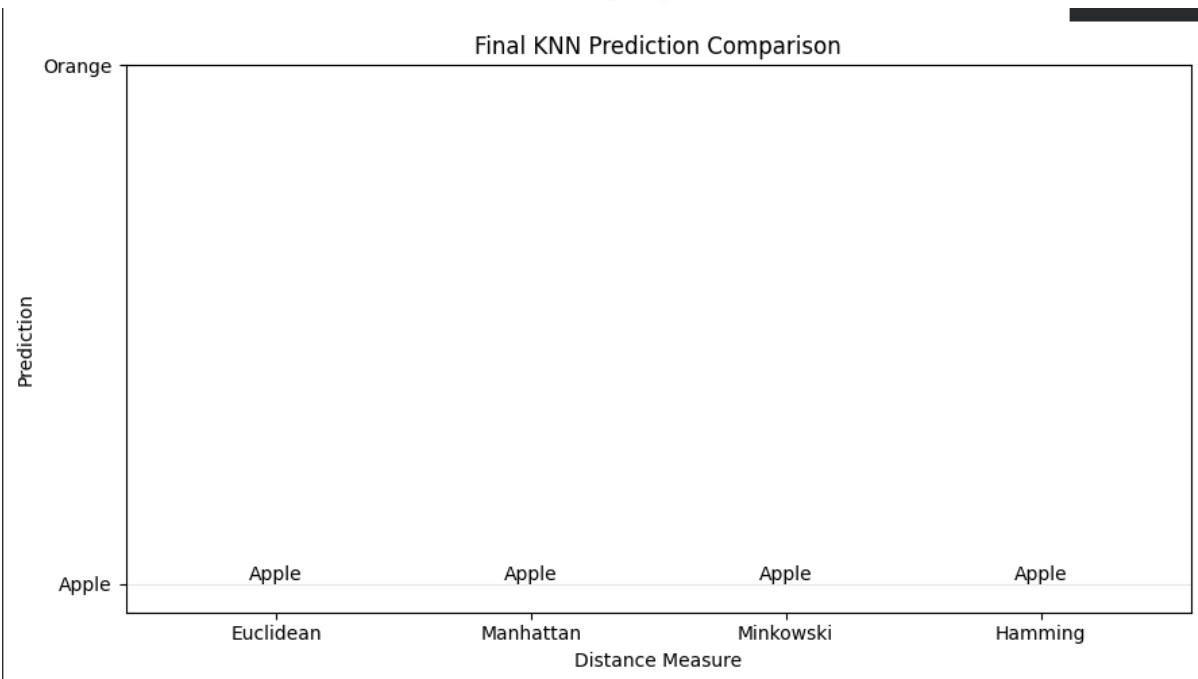
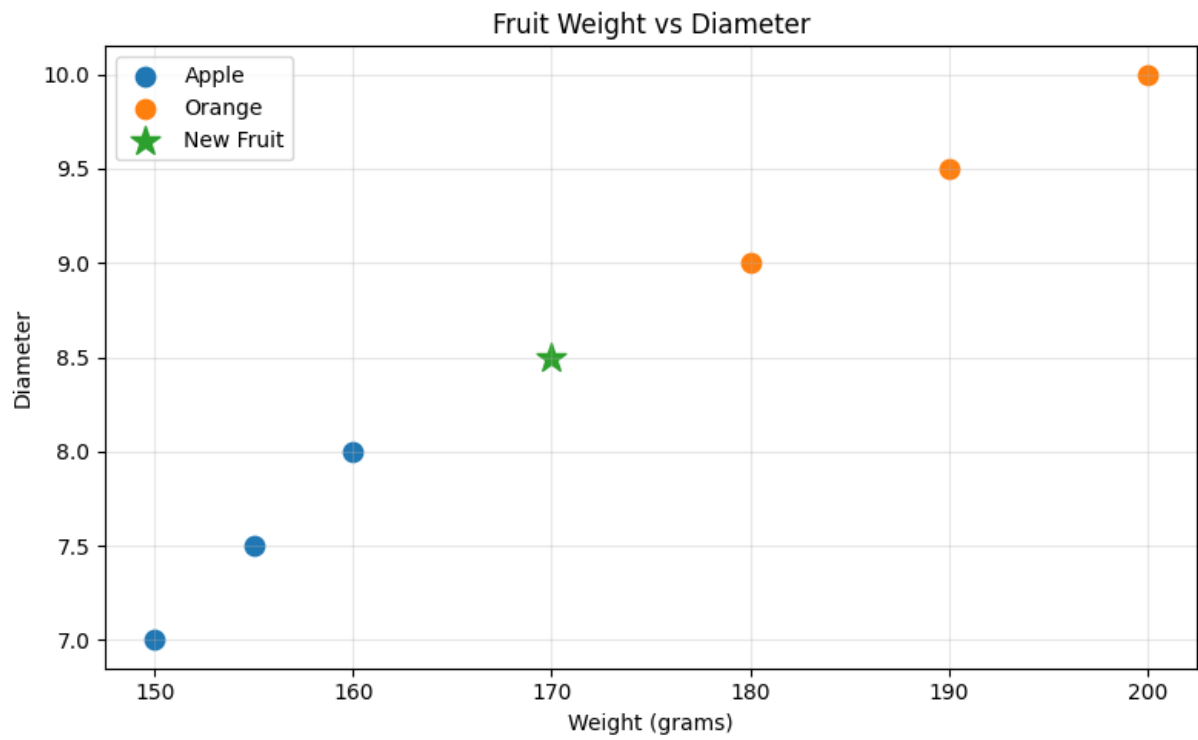
FINAL PREDICTIONS

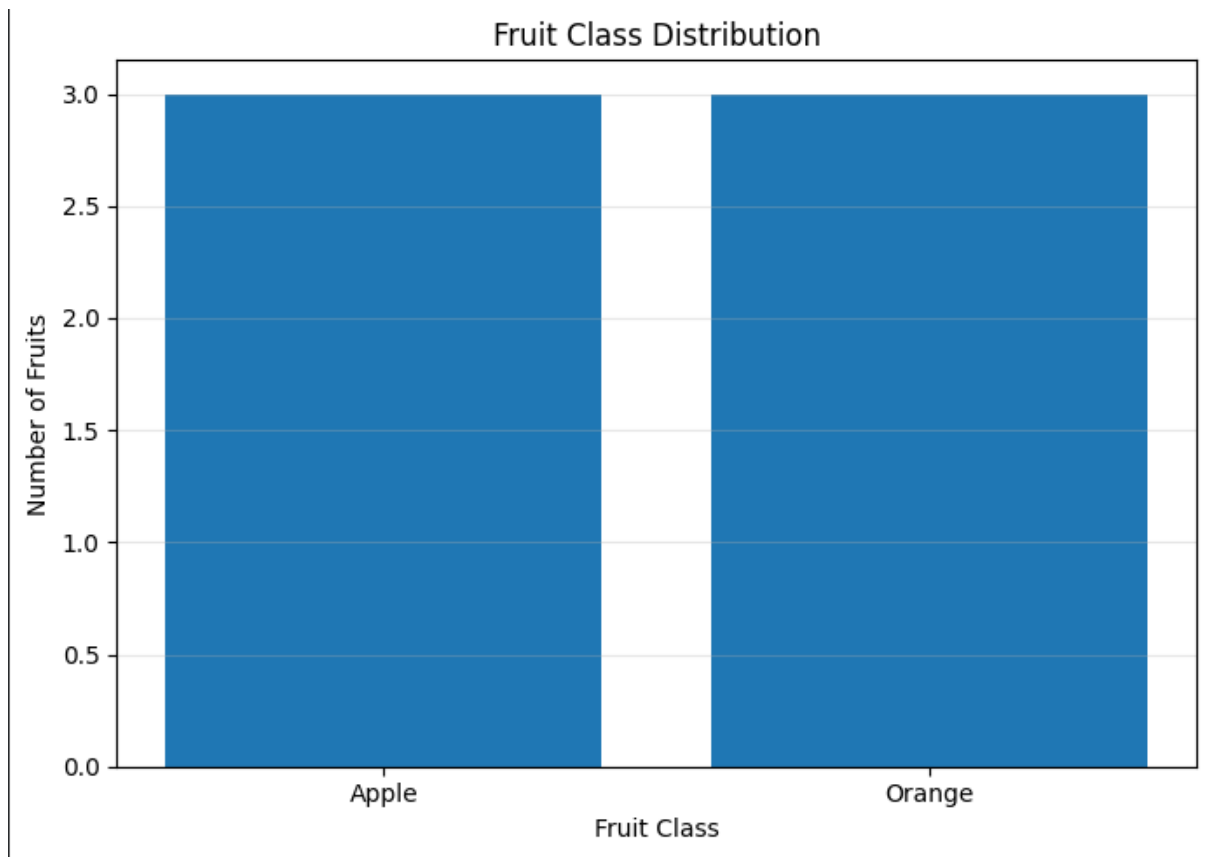
=====

Euclidean	: Apple
Manhattan	: Apple
Minkowski	: Apple
Hamming	: Apple









Question 5: House Price Category Prediction

A real estate company wants to classify houses as Low Price or High Price.

House	Area in sq.ft	Bedrooms	Parking Available	Near Main Road	Garden Available	Category
H1	800	2	0	0	0	Low Price
H2	1000	2	1	0	0	Low Price
H3	1500	3	1	1	1	High Price
H4	1800	4	1	1	1	High Price
H5	1200	3	0	1	0	Low Price
H6	2000	4	1	1	1	High Price

A new house has:

Area	Bedrooms	Parking Available	Near Main Road	Garden Available
1400	3	1	1	0

Using KNN with $K = 3$, classify the house as Low Price or High Price using Euclidean, Manhattan, Minkowski, and Hamming distance.

⑤ Question: - House Price Category Prediction

$$X = (1000, 3, 1, 1, 0)$$

$$k=3$$

Formula:-

$$D_E = \sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$

$$D_E = \sqrt{600^2 + 1^2 + 1^2 + 1^2 + 0^2} = 600.0025$$

$$H_2 = 400.0025$$

$$H_3 = 100.0025, H_4 = 400.0025, H_6 = 600.0025$$

Prediction = Low Price

2. Manhattan Distance

$$D_M = \sum |x_i - y_i|$$

$$H_1 = 600 + 1 + 1 + 0 + 1 = 603$$

$$H_2 = 402, H_3 = 100, H_4 = 402, H_5 = 201, H_6 = 602$$

Prediction = Low Price

3. Minkowski Distance ($p=3$)

$$D_M = \left(\sum |x_i - y_i|^3 \right)^{1/3}$$

$$H_1 = 600.0000, H_2 = 400.000, H_3 = 100, H_4 = 400, H_5 = 200, H_6 = 600$$

4. Hamming Distance

D_H = No. of different binary values

$$H_1 = 2, H_2 = 1, H_3 = 1, H_4 = 1, H_5 = 1, H_6 = 1$$

Prediction = High Price

Output:

HOUSE PRICE DATA

	House	Area	Bedrooms	Parking_Available	Near_Main_Road
0	H1	800	2	0	0
0					
1	H2	1000	2	1	0
0					
2	H3	1500	3	1	1
1					
3	H4	1800	4	1	1
1					
4	H5	1200	3	0	1
0					
5	H6	2000	4	1	1
1					

	Category
0	Low Price
1	Low Price
2	High Price
3	High Price
4	Low Price
5	High Price

=====

DISTANCE CALCULATIONS

=====

House	Category	Euclidean	Manhattan	Minkowski	Hamming
H1	Low Price	600.0025	603	600.0	2
H2	Low Price	400.0025	402	400.0	1
H3	High Price	100.0050	101	100.0	1
H4	High Price	400.0025	402	400.0	1
H5	Low Price	200.0025	201	200.0	1
H6	High Price	600.0017	602	600.0	1

=====

EUCLIDEAN DISTANCE

=====

3 Nearest Neighbours:

House	Category	Distance
H3	High Price	100.0050
H5	Low Price	200.0025
H4	High Price	400.0025

Majority Voting:

Category
High Price 2
Low Price 1

Name: count, dtype: int64

Prediction: High Price

=====

MANHATTAN DISTANCE

=====

3 Nearest Neighbours:

House	Category	Distance
-------	----------	----------

H3 High Price	101
H5 Low Price	201
H4 High Price	402

Majority Voting:

Category

High Price 2

Low Price 1

Name: count, dtype: int64

Prediction: High Price

=====

MINKOWSKI DISTANCE (p = 3)

=====

3 Nearest Neighbours:

House	Category	Distance
-------	----------	----------

H3 High Price	100.000033
---------------	------------

H5 Low Price	200.000008
--------------	------------

H4 High Price	400.000004
---------------	------------

Majority Voting:

Category

High Price 2

Low Price 1

Name: count, dtype: int64

Prediction: High Price

=====

HAMMING DISTANCE

=====

3 Nearest Neighbours:

House	Category	Distance
-------	----------	----------

H2 Low Price	1
--------------	---

H3 High Price	1
---------------	---

H4 High Price	1
---------------	---

Majority Voting:

Category

High Price 2

Low Price 1

Name: count, dtype: int64

Prediction: High Price

=====

FINAL PREDICTIONS

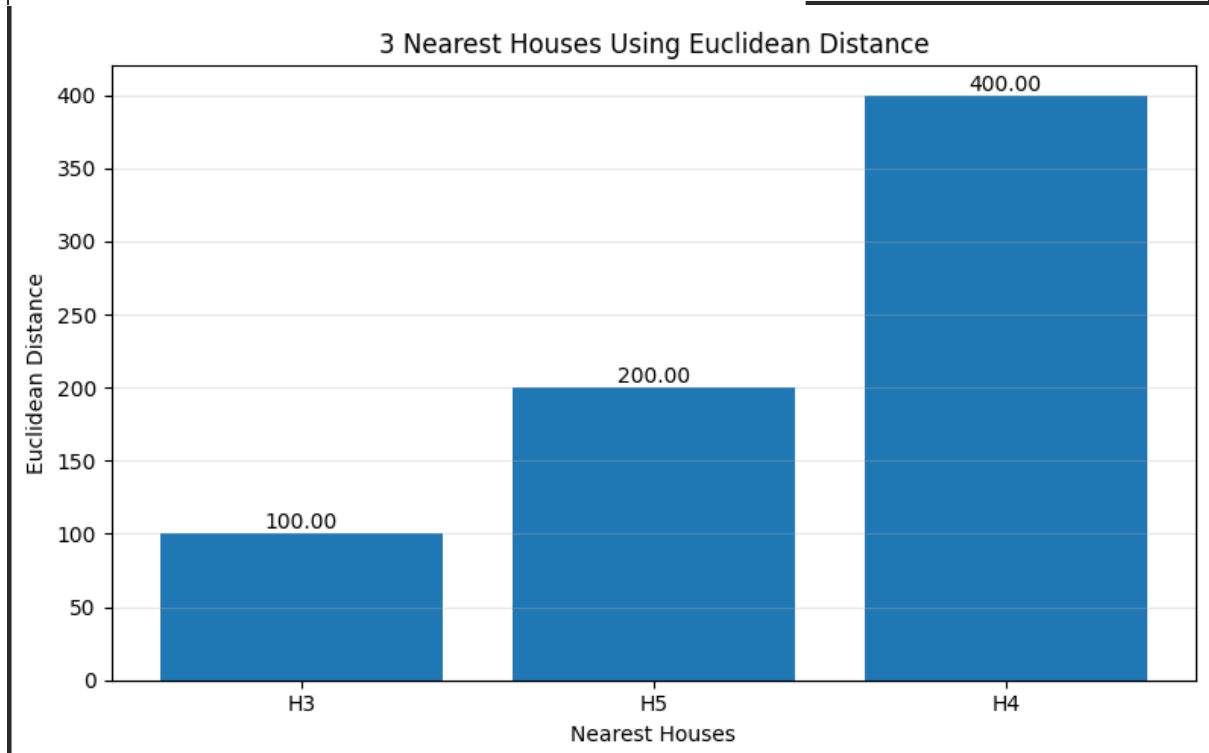
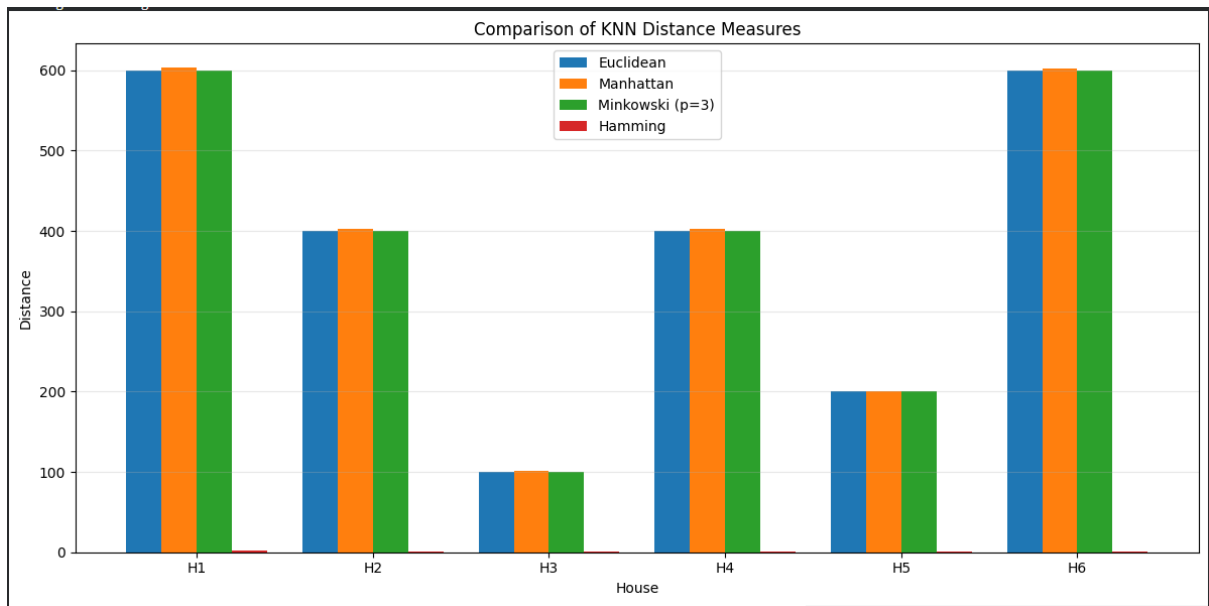
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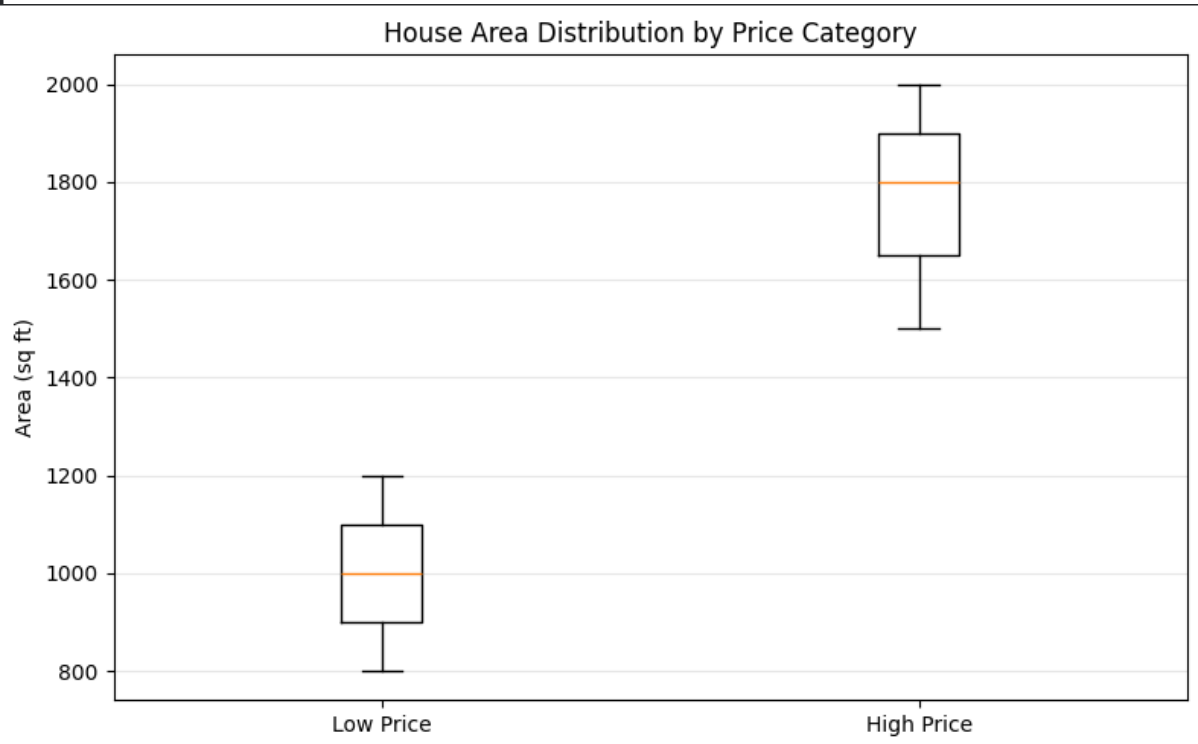
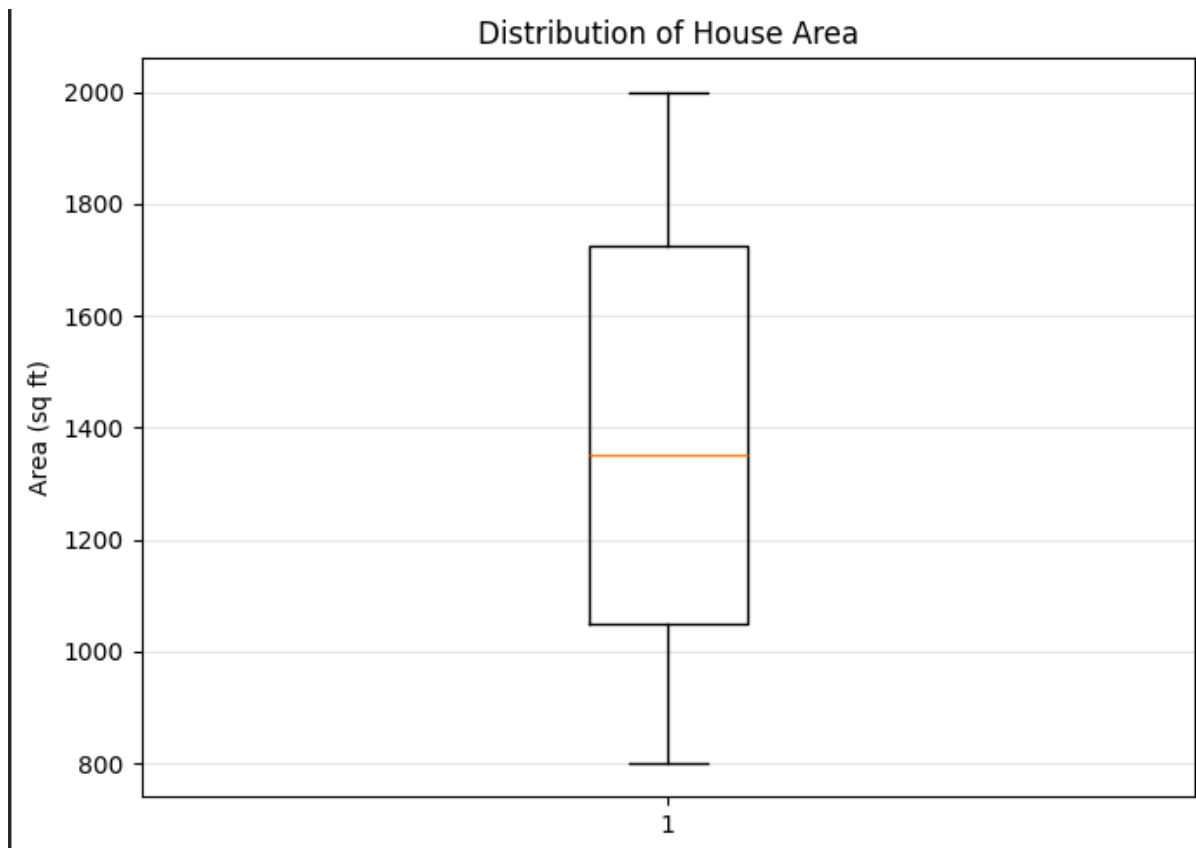
Euclidean : High Price

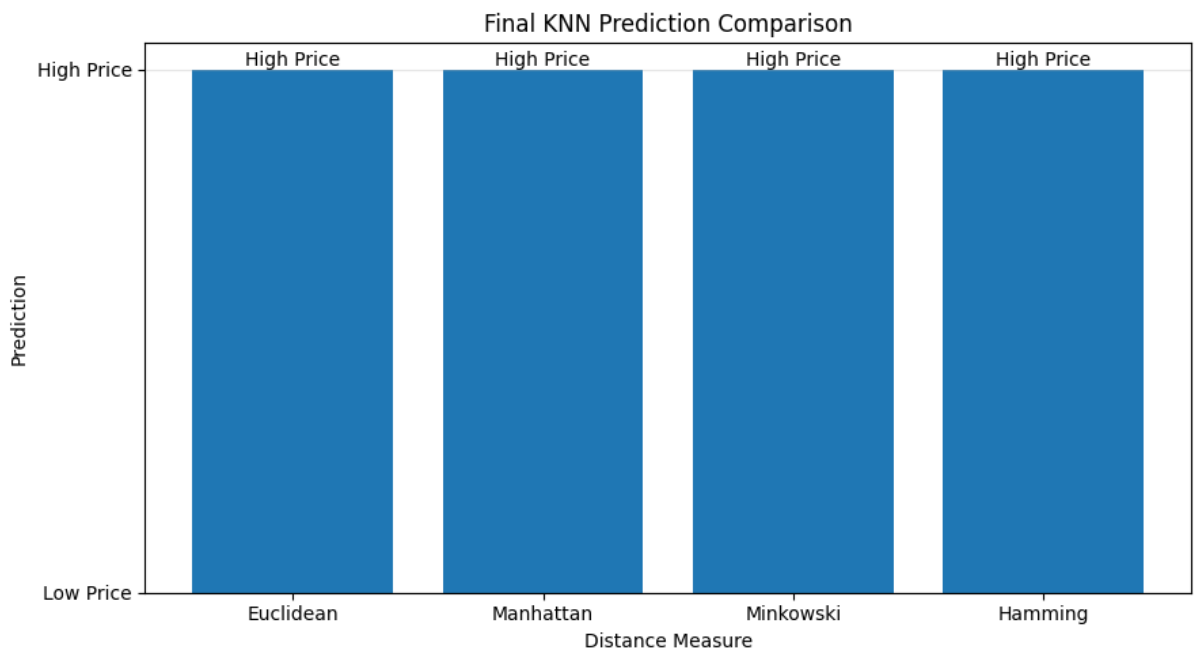
Manhattan : High Price

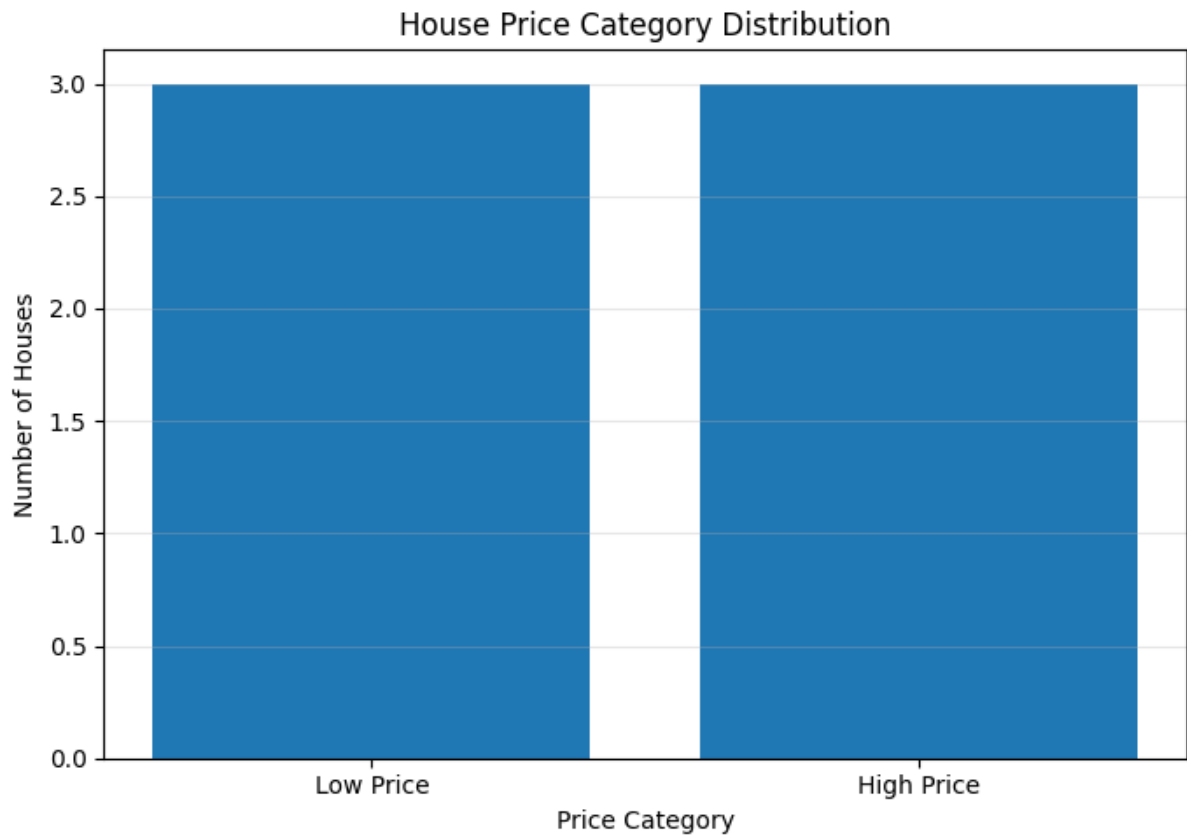
Minkowski : High Price

Hamming : High Price









Question 6: Loan Approval Prediction

A bank wants to predict whether a loan application should be Approved or Rejected.

Applicant	Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default	Class
A1	25000	580	0	1	1	Rejected
A2	30000	600	0	1	0	Rejected
A3	60000	750	1	0	0	Approved
A4	70000	800	1	0	0	Approved
A5	40000	650	0	1	0	Rejected
A6	80000	820	1	0	0	Approved

A new applicant has:

Monthly Income	Credit Score	Govt Job	Existing Loan	Past Default
55000	700	1	0	0

Using KNN with $K = 3$, classify the loan application as Approved or Rejected using all four distance measures.

Question 6: Loan Approval Prediction :-

$$X = (55000, 700, 1, 0, 0)$$

$$k=3$$

Eucclidean :-

$$\text{Formula :- } D_E = \sqrt{\sum (x_i - y_i)^2}$$

$$A_1 = \sqrt{30000^2 + 120^2 + 1^2 + 1^2 + 1^2} = 30000.2400$$

$$A_2 = 25000 \cdot 2000, A_3 = 5000 \cdot 2500, A_4 = 15000 \cdot 3338, A_5 = 11000$$

$$A_6 = 25000 \cdot 2000$$

Prediction = Approved

Manhattan Distance

$$D_M = \sum |x_i - y_i|$$

$$A_1 = 30000 + 120 + 1 + 1 + 1$$

$$A_1 = 20123$$

$$A_2 = 25102, A_3 = 5010, A_4 = 15100, A_5 = 11052, A_6 = 25120$$

Prediction = Approved

Minkowski Distance : (p=3)

$$D_M = (\sum |x_i - y_i|^3)^{1/3}$$

$$A_1 = 30000 \cdot 0006, A_2 = 25000 \cdot 0005, A_3 = 5000 \cdot 0019$$

Prediction = Approved

Hamming Distance

$$\text{Formula :- } D_H = \text{No. of different binary values} \\ (1, 0, 0)$$

$$A_1 \rightarrow 3, A_2 \rightarrow 2, A_3 \rightarrow 0, A_4 \rightarrow 0, A_5 \rightarrow 2,$$

$$A_6 \rightarrow 0$$

Prediction = Approved

Output:

LOAN APPROVAL DATA

	Applicant	Monthly_Income	Credit_Score	Govt_Job	Existing_Loan	\
0	A1	25000	580	0	1	
1	A2	30000	600	0	1	
2	A3	60000	750	1	0	
3	A4	70000	800	1	0	
4	A5	40000	650	0	1	
5	A6	80000	820	1	0	

	Past_Default	Class
0	1	Rejected
1	0	Rejected
2	0	Approved
3	0	Approved
4	0	Rejected
5	0	Approved

DISTANCE CALCULATIONS

Applicant	Class	Euclidean	Manhattan	Minkowski	Hamming
A1	Rejected	30000.2400	30123	30000.0006	3
A2	Rejected	25000.2000	25102	25000.0005	2
A3	Approved	5000.2500	5050	5000.0017	0
A4	Approved	15000.3333	15100	15000.0015	0
A5	Rejected	15000.0834	15052	15000.0002	2
A6	Approved	25000.2880	25120	25000.0009	0

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Applicant	Class	Distance
A3	Approved	5000.249994
A5	Rejected	15000.083400
A4	Approved	15000.333330

Majority Voting:

Class

Approved 2

Rejected 1

Name: count, dtype: int64

Prediction: Approved

MANHATTAN DISTANCE

3 Nearest Neighbours:

Applicant	Class	Distance
A3	Approved	5050
A5	Rejected	15052
A4	Approved	15100

Majority Voting:

Class

Approved 2

Rejected 1
Name: count, dtype: int64

Prediction: Approved

=====
MINKOWSKI DISTANCE (p = 3)
=====

3 Nearest Neighbours:
Applicant Class Distance
A3 Approved 5000.001667
A5 Rejected 15000.000185
A4 Approved 15000.001481

Majority Voting:
Class
Approved 2
Rejected 1
Name: count, dtype: int64

Prediction: Approved

=====
HAMMING DISTANCE
=====

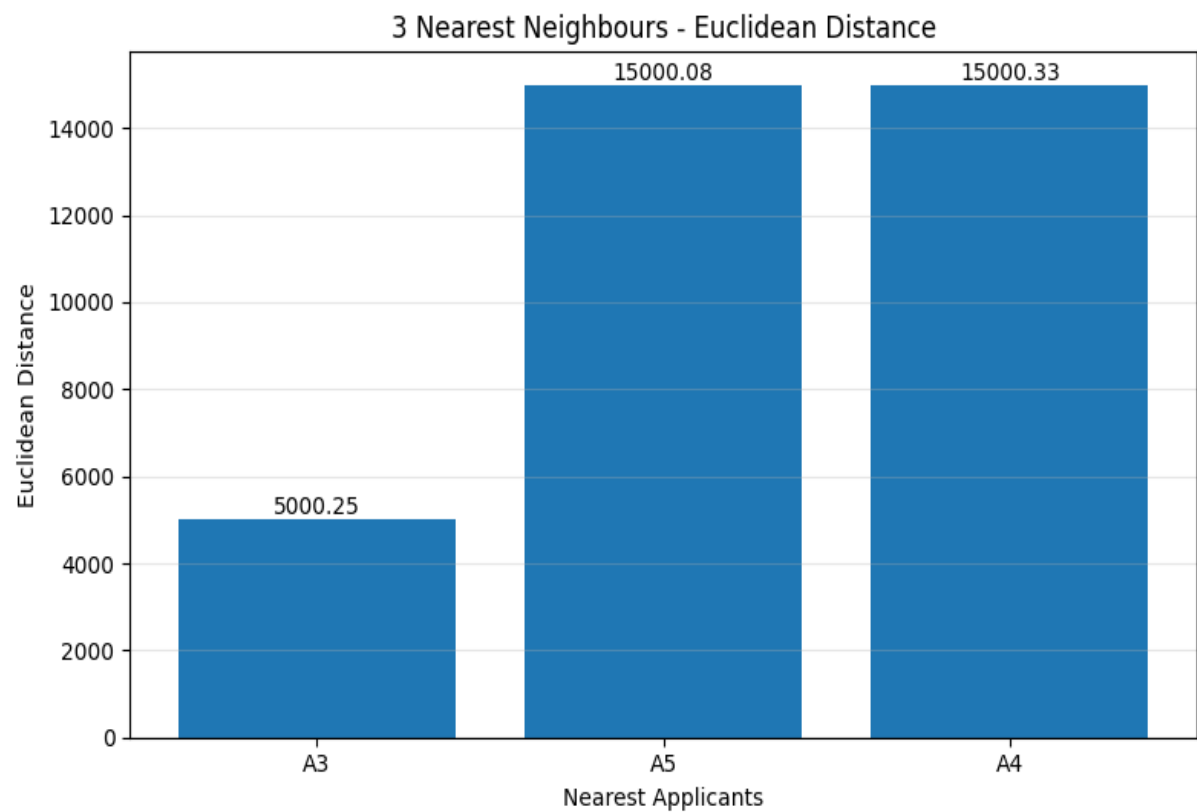
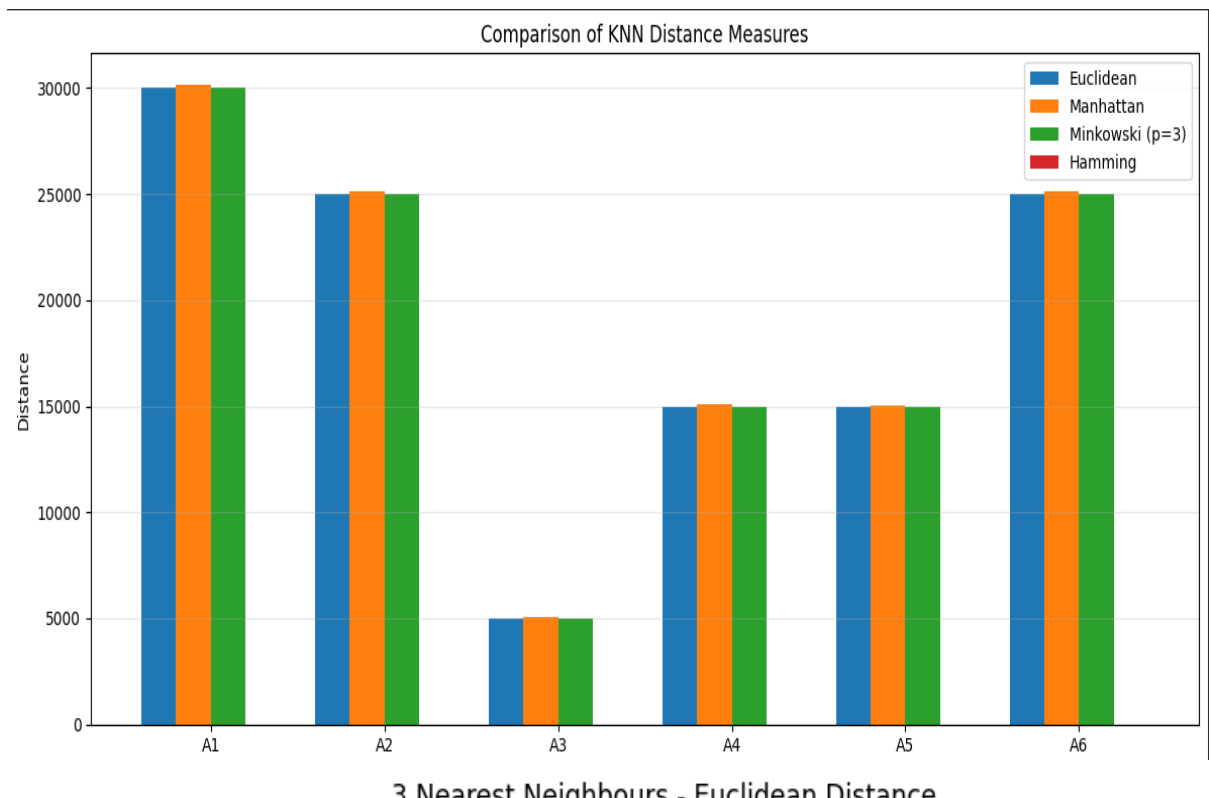
3 Nearest Neighbours:
Applicant Class Distance
A4 Approved 0
A3 Approved 0
A6 Approved 0

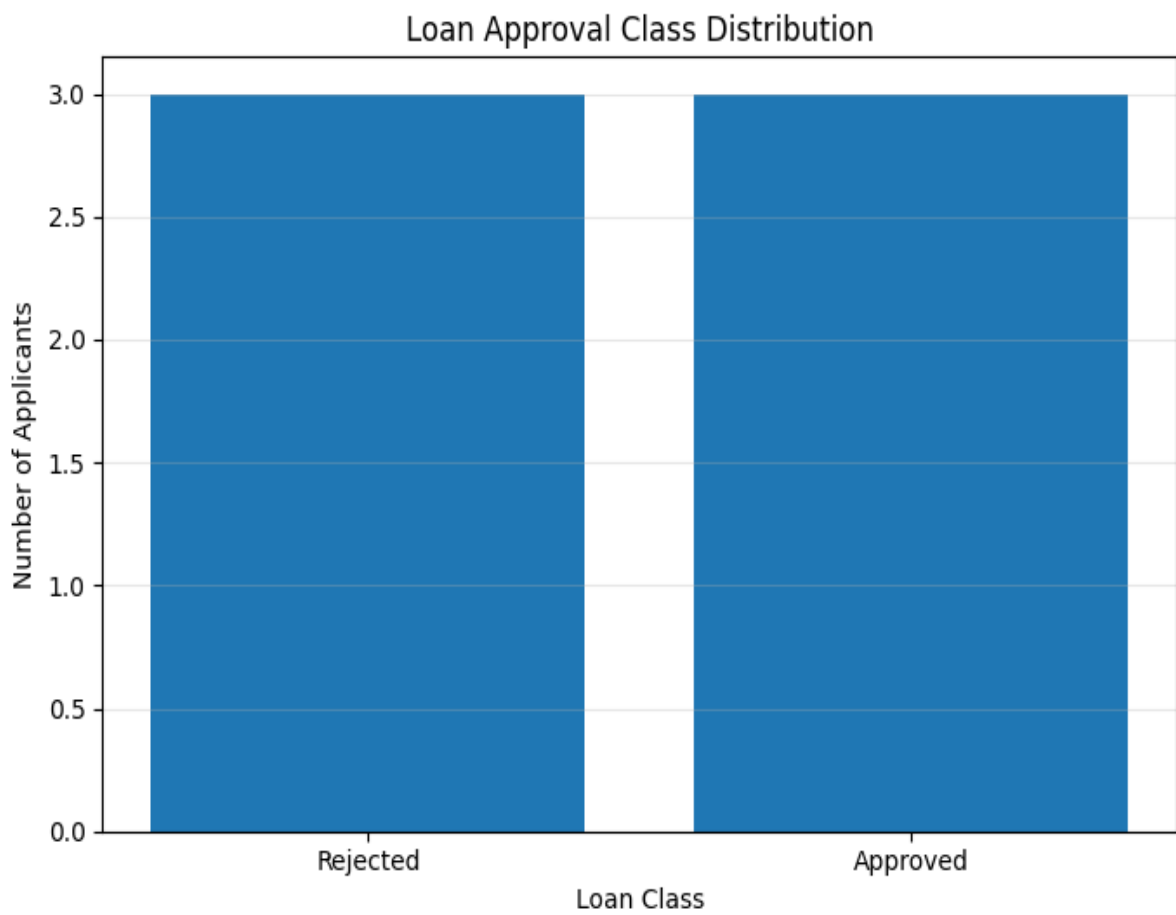
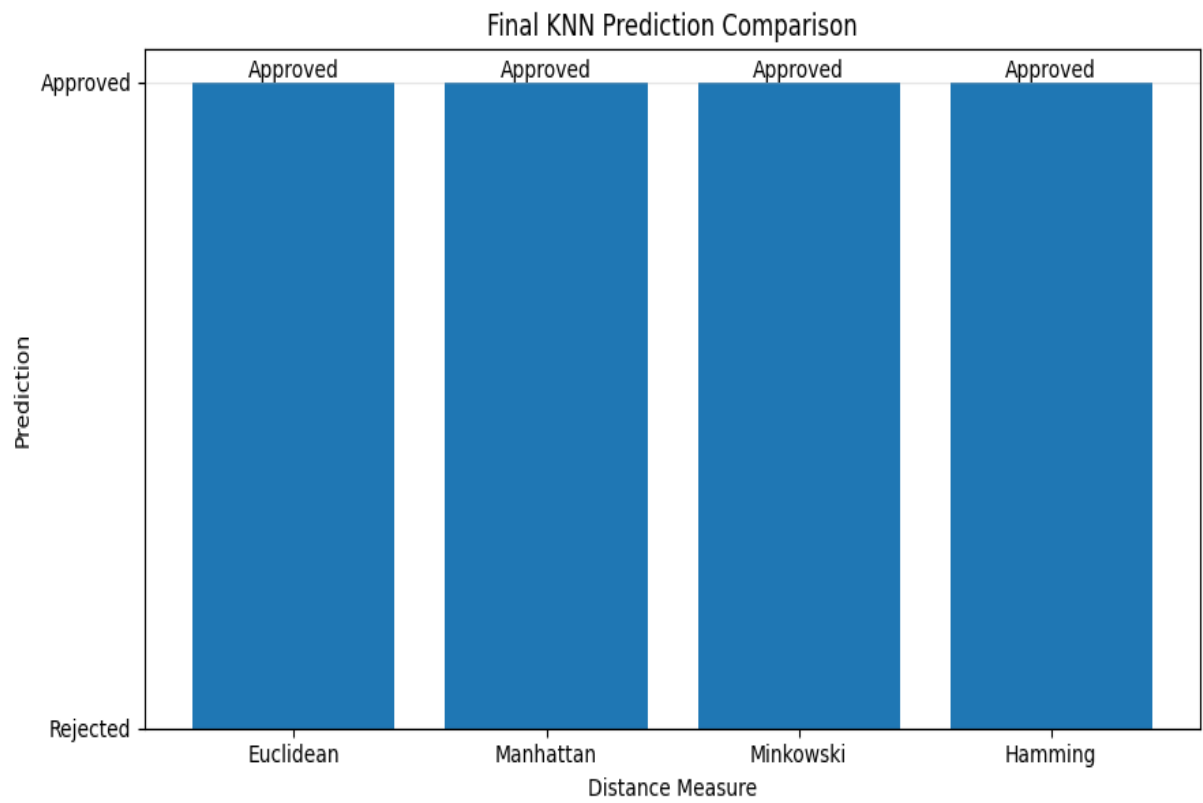
Majority Voting:
Class
Approved 3
Name: count, dtype: int64

Prediction: Approved

=====
FINAL PREDICTIONS
=====

Euclidean : Approved
Manhattan : Approved
Minkowski : Approved
Hamming : Approved





Question 7: Movie Preference Classification

A movie platform wants to classify a user as an Action Lover or Comedy Lover.

User	Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom	Class
U1	5	1	1	0	0	Action Lover
U2	4	2	1	0	0	Action Lover
U3	1	5	0	1	1	Comedy Lover
U4	2	4	0	1	1	Comedy Lover
U5	5	2	1	0	0	Action Lover
U6	1	4	0	1	1	Comedy Lover

A new user has:

Action Rating	Comedy Rating	Watched Superhero Movie	Watched Stand-up Show	Watched Sitcom
3	3	1	1	0

Using KNN with $K = 3$, classify the new user as Action Lover or Comedy Lover using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 1: Movie Preference Classification

$$X = (3, 3, 1, 1, 0)$$

$$k=3$$

1. Euclidean Distance :-

$$D_e = \sqrt{\sum (x_i - y_i)^2}$$

$$U_1 = D = \sqrt{(3-3)^2 + (3-1)^2 + (1-1)^2 + (1-0)^2 + (0-0)^2}$$

$$= \sqrt{4+4+0+1+0} = \sqrt{9}$$

$$D(U_1) = 3$$

Action Lover

$$U_2 = 1.7221, U_3 = 3.1623, U_4 = 2, U_5 = 2.9915, U_6 = 2.6858$$

2. Manhattan Distance :-

$$D_M = \sum |x_i - y_i|$$

$$U_1 = 2+2+0+1+0 = 5$$

$$D_1 = 5$$

Action Lover

$$U_2 = 3, U_3 = 6, U_4 = 4, U_5 = 4, U_6 = 5$$

3. Minkowski Distance ($p=3$)

$$D_M = (\sum |x_i - y_i|^3)^{1/3}$$

$$U_1 = D = (2^3 + 2^3 + 1^3)^{1/3} = 9^{1/3}$$

$$D(U_1) = 2.0813$$

Action Lover

$$U_2 = 1.4422, U_3 = 2.6209, U_4 = 1.5874, U_5 = 2.1544, U_6 = 2.2290$$

4. Hamming Distance :-

$$X_B = (1, 1, 0)$$

$$D_H = \text{No. of different binary values}$$

$$U_1 = (1, 0, 0) \rightarrow 1$$

$$U_2 = 1, U_3 = 2, U_4 = 2, U_5 = 1, U_6 = 2$$

Action Lover

Output:

MOVIE PREFERENCE DATA

	User	Action_Rating	Comedy_Rating	Watched_Superhero	Watched_Standup	\
0	U1	5	1	1	0	
1	U2	4	2	1	0	
2	U3	1	5	0	1	
3	U4	2	4	0	1	
4	U5	5	2	1	0	
5	U6	1	4	0	1	

	Watched_Sitcom	Class
0	0	Action Lover
1	0	Action Lover
2	1	Comedy Lover
3	1	Comedy Lover
4	0	Action Lover
5	1	Comedy Lover

DISTANCE CALCULATIONS

User	Class	Euclidean	Manhattan	Minkowski	Hamming
U1	Action Lover	3.0000	5	2.5713	1
U2	Action Lover	1.7321	3	1.4422	1
U3	Comedy Lover	3.1623	6	2.6207	2
U4	Comedy Lover	2.0000	4	1.5874	2
U5	Action Lover	2.4495	4	2.1544	1
U6	Comedy Lover	2.6458	5	2.2240	2

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

User	Class	Distance
U2	Action Lover	1.732051
U4	Comedy Lover	2.000000
U5	Action Lover	2.449490

Majority Voting:

Class

Action Lover 2

Comedy Lover 1

Name: count, dtype: int64

Prediction: Action Lover

MANHATTAN DISTANCE

3 Nearest Neighbours:

User	Class	Distance
U2	Action Lover	3
U4	Comedy Lover	4
U5	Action Lover	4

Majority Voting:

Class

Action Lover 2

Comedy Lover 1
Name: count, dtype: int64

Prediction: Action Lover

=====
MINKOWSKI DISTANCE (p = 3)
=====

3 Nearest Neighbours:
User Class Distance
U2 Action Lover 1.442250
U4 Comedy Lover 1.587401
U5 Action Lover 2.154435

Majority Voting:
Class
Action Lover 2
Comedy Lover 1
Name: count, dtype: int64

Prediction: Action Lover

=====
HAMMING DISTANCE
=====

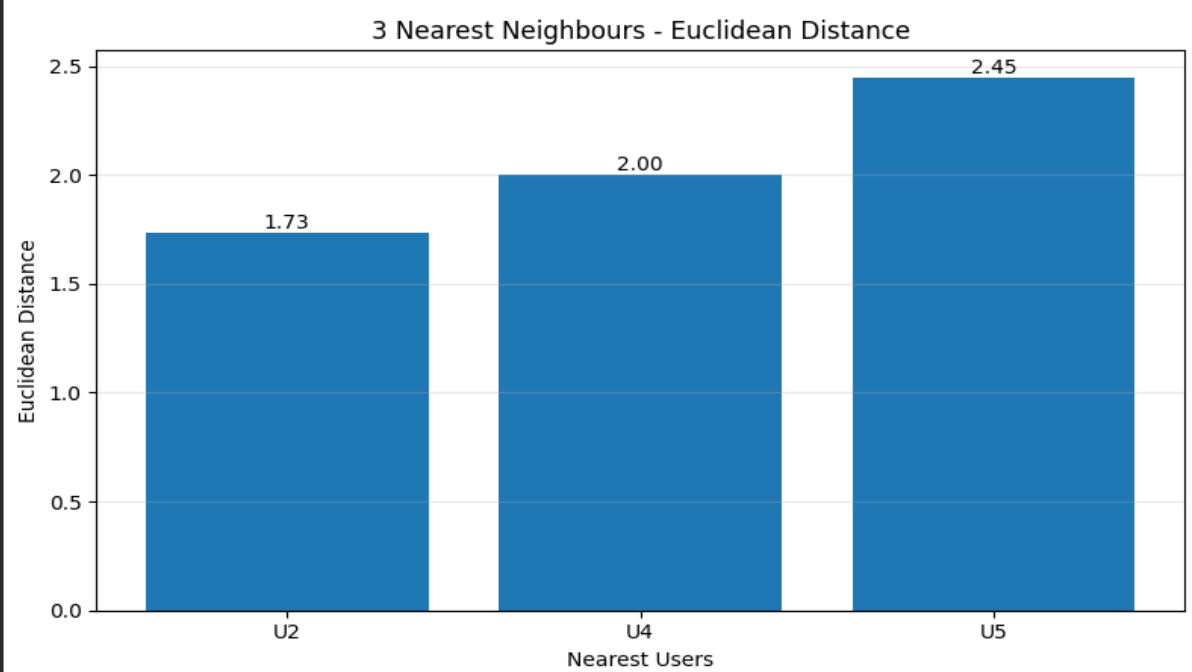
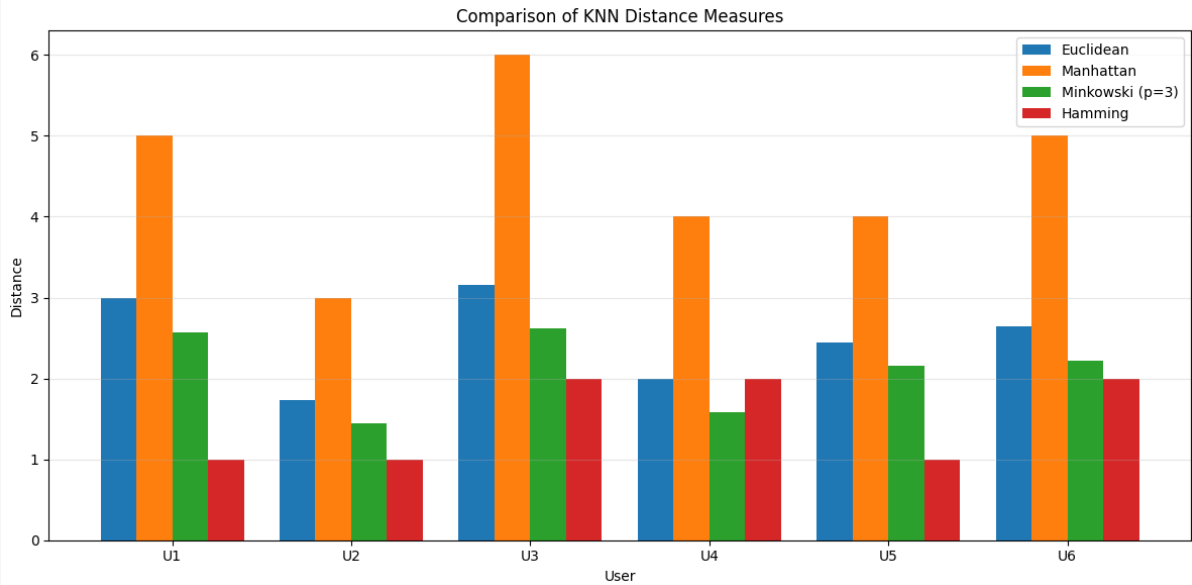
3 Nearest Neighbours:
User Class Distance
U1 Action Lover 1
U2 Action Lover 1
U5 Action Lover 1

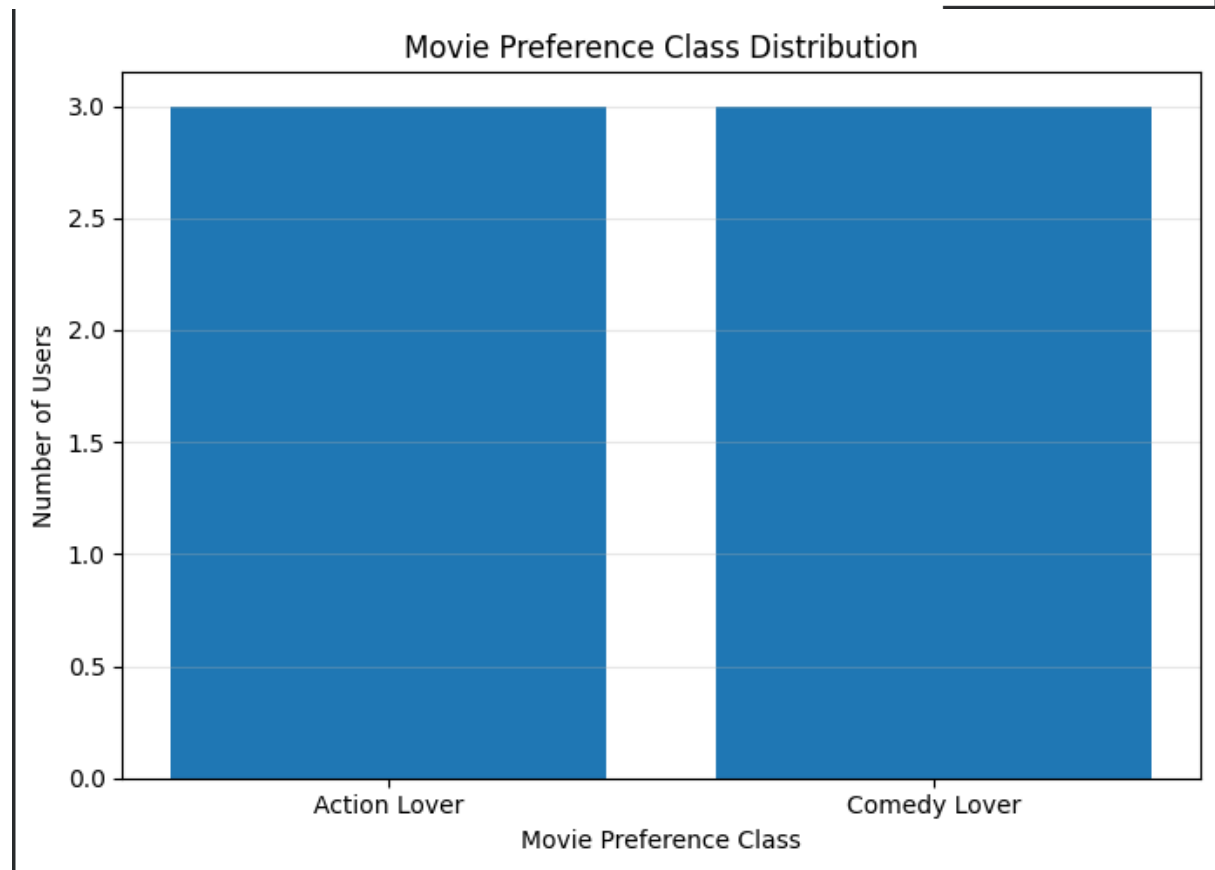
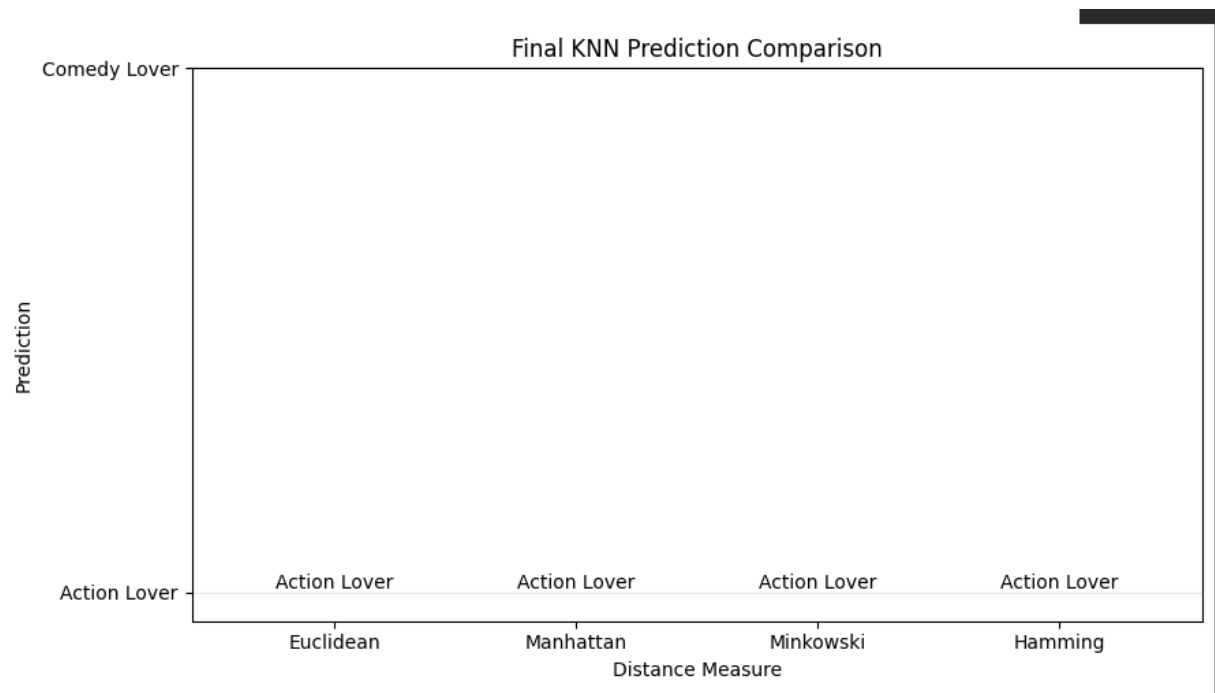
Majority Voting:
Class
Action Lover 3
Name: count, dtype: int64

Prediction: Action Lover

=====
FINAL PREDICTIONS
=====

Euclidean : Action Lover
Manhattan : Action Lover
Minkowski : Action Lover
Hamming : Action Lover





Question 8: Crop Disease Detection

An agriculture research center wants to classify a plant as Healthy or Diseased.

Plant	Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting	Class
L1	2	80	0	0	0	Healthy
L2	3	75	0	0	0	Healthy
L3	12	40	1	1	1	Diseased
L4	15	35	1	1	1	Diseased
L5	5	70	0	0	1	Healthy
L6	14	38	1	1	1	Diseased

A new plant has:

Leaf Spot Count	Leaf Color Index	Yellow Leaves	White Fungus	Wilting
10	50	1	1	0

Using KNN with $K = 3$, classify the plant as Healthy or Diseased using all four distance measures.

Question 8: Crop Disease Detection

$X = (10, 50, 1, 1, 0)$

$k=3$

1. Euclidean Distance

$$D_E = \sqrt{\sum (x_i - y_i)^2}$$

$$L_1 = \sqrt{5^2 + 30^2 + 1^2 + 1^2} = 31.368$$

$$L_2 = 25.9615, L_3 = 10.2970, L_4 = 15.8420, L_5 = 20.2985, L_6 = 12.4826$$

Prediction = Diseased

2. Manhattan Distance

$$D_M = \sum |x_i - y_i|$$

$$L_1 = (8 + 30 + 1 + 1 + 0) = 40$$

$$L_2 = 34, L_3 = 13, L_4 = 21, L_5 = 22, L_6 = 19$$

Prediction = Diseased

3. Minkowski Distance ($p=3$)

$$D_M = (\sum |x_i - y_i|^3)^{1/3}$$

$$L_1 = (5^3 + 30^3 + 1^3 + 1^3)^{1/3} = 30.0187$$

$$L_2 = 25.0157, L_3 = 10.0052, L_4 = 15.0185, L_5 = 20.0164, L_6 = 12.019$$

Prediction = Diseased

4. Hamming Distance
(Yellow leaves, White Fungus, Wilting)

New plant: $X_P = (1, 1, 0)$

D_H = No. of different binary values

$$L_1 \rightarrow (0, 0, 0) \rightarrow 2, L_2 \rightarrow (0, 0, 0) \rightarrow 2, L_3 \rightarrow (1, 1, 1) \rightarrow 3$$

$$L_4 \rightarrow 1, L_5 \rightarrow 3, L_6 \rightarrow 1$$

Prediction = Diseased

Output:

CROP DISEASE DATA

	Plant	Leaf_Spot_Count	Leaf_Color_Index	Yellow_Leaves	White_Fungus	\
0	L1	2	80	0	0	
1	L2	3	75	0	0	
2	L3	12	40	1	1	
3	L4	15	35	1	1	
4	L5	5	70	0	0	
5	L6	14	38	1	1	

	Wilting	Class
0	0	Healthy
1	0	Healthy
2	1	Diseased
3	1	Diseased
4	1	Healthy
5	1	Diseased

DISTANCE CALCULATIONS

Plant	Class	Euclidean	Manhattan	Minkowski	Hamming
L1	Healthy	31.0805	40	30.1892	2
L2	Healthy	26.0000	34	25.1827	2
L3	Diseased	10.2470	13	10.0299	1
L4	Diseased	15.8430	21	15.1844	1
L5	Healthy	20.6882	28	20.1061	3
L6	Diseased	12.6886	17	12.1486	1

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Plant	Class	Distance
L3	Diseased	10.246951
L6	Diseased	12.688578
L4	Diseased	15.842980

Majority Voting:

Class

Diseased 3

Name: count, dtype: int64

Prediction: Diseased

MANHATTAN DISTANCE

3 Nearest Neighbours:

Plant	Class	Distance
L3	Diseased	13
L6	Diseased	17
L4	Diseased	21

Majority Voting:

Class

Diseased 3

Name: count, dtype: int64

Prediction: Diseased

```
=====
MINKOWSKI DISTANCE (p = 3)
=====
```

3 Nearest Neighbours:

Plant	Class	Distance
L3	Diseased	10.029910
L6	Diseased	12.148615
L4	Diseased	15.184391

Majority Voting:
Class
Diseased 3
Name: count, dtype: int64

Prediction: Diseased

```
=====
HAMMING DISTANCE
=====
```

3 Nearest Neighbours:

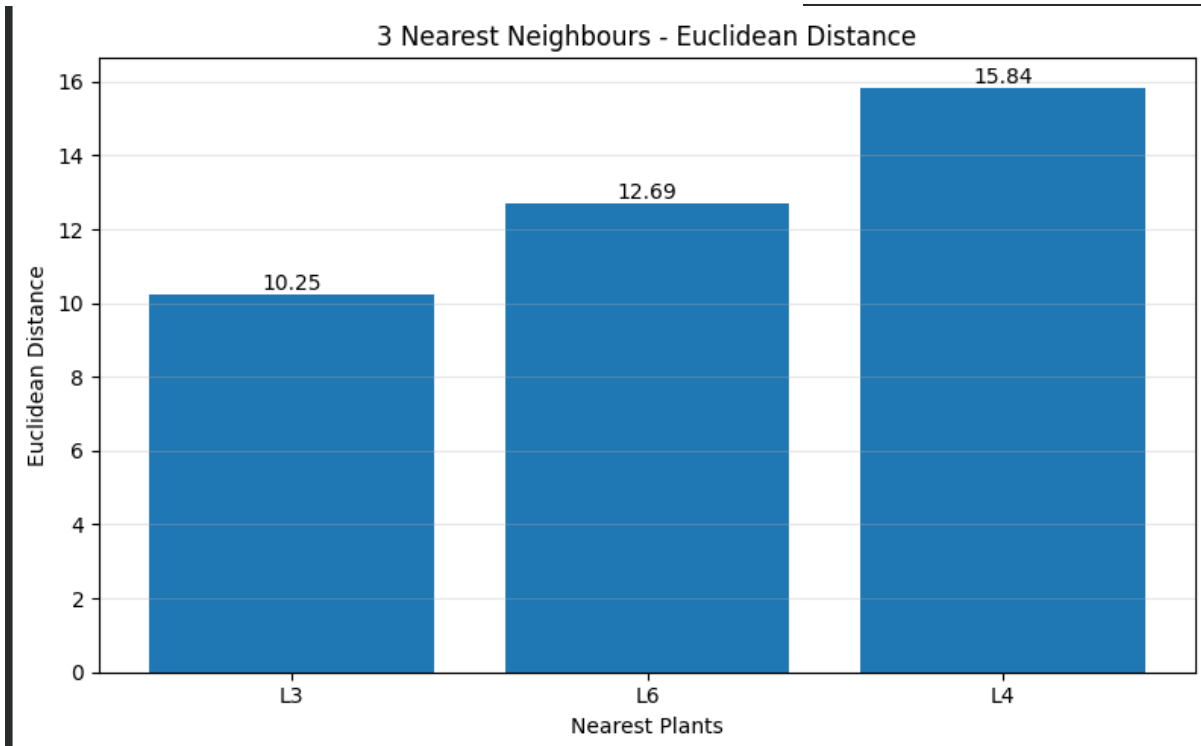
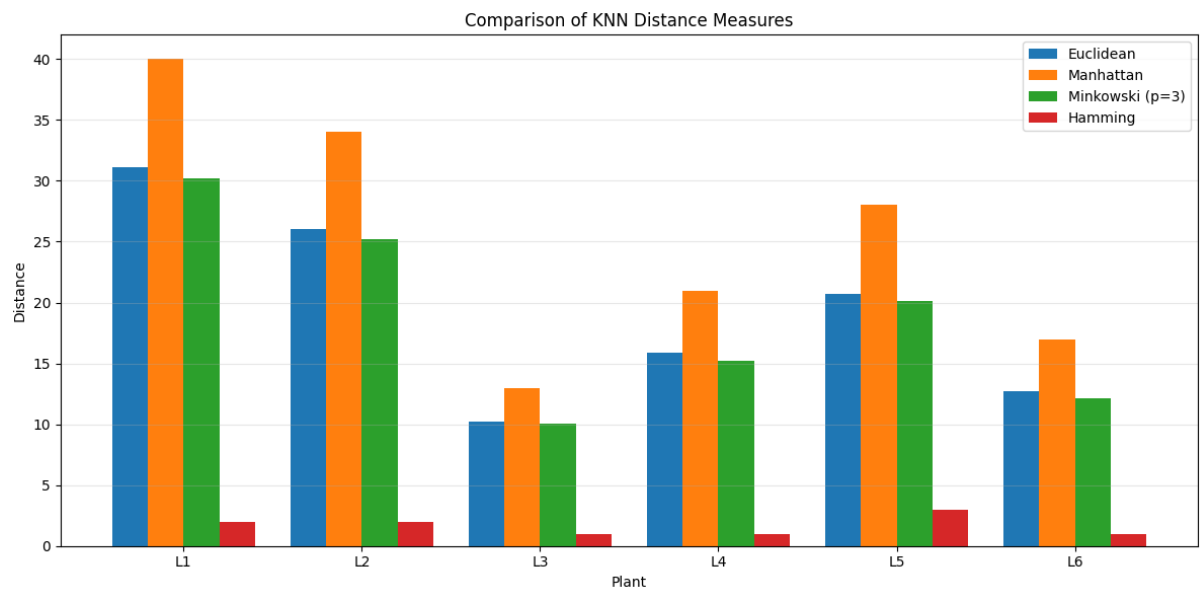
Plant	Class	Distance
L4	Diseased	1
L3	Diseased	1
L6	Diseased	1

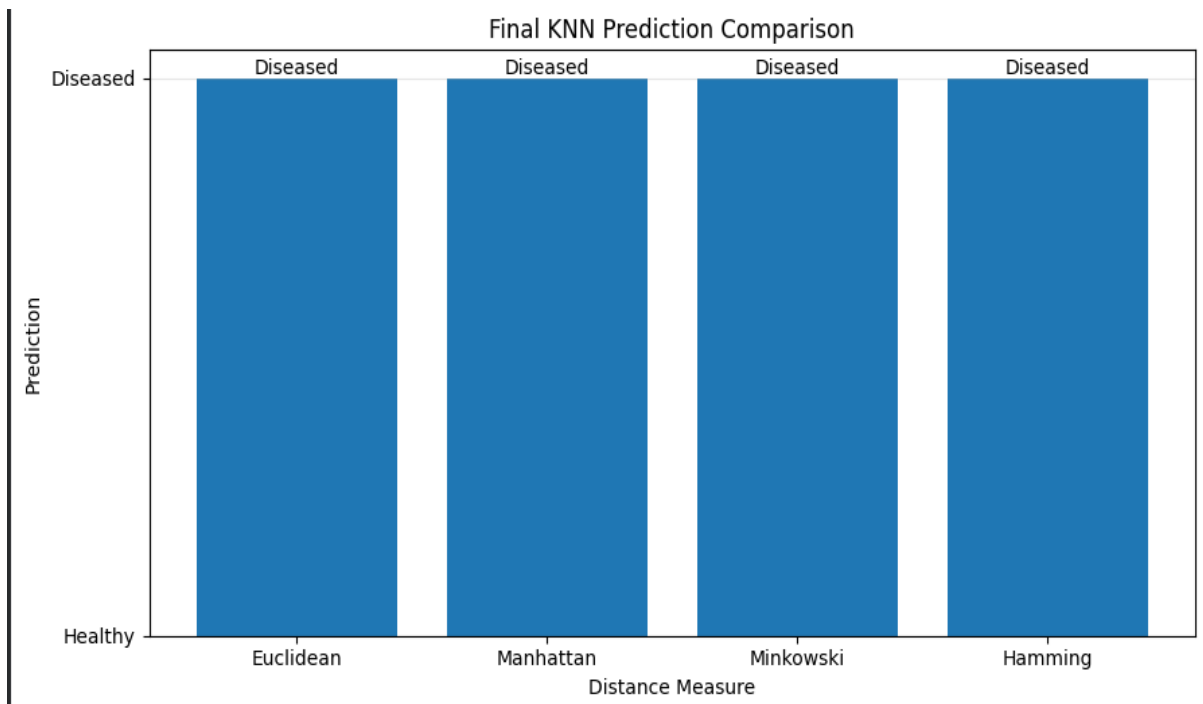
Majority Voting:
Class
Diseased 3
Name: count, dtype: int64

Prediction: Diseased

```
=====
FINAL PREDICTIONS
=====
```

Euclidean	: Diseased
Manhattan	: Diseased
Minkowski	: Diseased
Hamming	: Diseased





Question 9: Network Intrusion Detection

A cybersecurity system wants to classify network activity as Normal or Attack.

Record	Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP	Class
N1	20	0	0	0	0	Normal
N2	30	1	0	0	0	Normal
N3	100	5	1	1	1	Attack
N4	120	6	1	1	1	Attack
N5	40	1	0	0	0	Normal
N6	110	7	1	1	1	Attack

A new network record has:

Packets per Second	Failed Login Attempts	Unknown Port Used	Admin Access Tried	Blacklisted IP
90	4	1	1	0

Using KNN with $K = 3$, classify the network activity as Normal or Attack using Euclidean, Manhattan, Minkowski, and Hamming distance.

Question 9: Network Intrusion Detection

$X = (90, 4, 1, 1, 0)$

where $k=3$

Binary features (Unknown Port Used, Admin Access Tried, Blacklisted IP)

1. Euclidean Distance

$$D_E = \sqrt{\sum (x_i - y_i)^2}$$

$N_1 = \sqrt{70^2 + 4^2 + 1^2 + 1^2 + 0^2} = 70.1265$
 $N_2 = 60.0916, N_3 = 10.0995, N_4 = 30.0132, N_5 = 50.1097, N_6 = 20.79$

Prediction = Attack

2. Manhattan Distance

$$D_M = \sum |x_i - y_i|$$

$N_1 = 70 + 4 + 1 + 1 + 0 = 76$
 $N_2 = 60 + 3 + 1 + 1 + 0 = 65$
 $N_3 = 12, N_4 = 23$
 $N_5 = 55, N_6 = 24$

Prediction = Attack

3. Minkowski Distance ($p=3$) :-

$$D_M = \left(\sum |x_i - y_i|^3 \right)^{1/3}$$

$$N_1 = (70^3 + 9^3 + 1^3 + 1^3)^{1/3} = 70.0001$$

$$N_2 = 60.0000, N_3 = 10.0067, N_4 = 30.0033, N_5 = 50.0012, N_6 = 20.0025$$

Prediction = Attack

4. Hamming Distance

$$X_B = (1, 1, 0)$$

D_H = No. of different binary values

$$N_1 \rightarrow (0, 0, 0), N_2 \rightarrow (0, 0, 0), N_3 \rightarrow (1, 1, 1), N_4 \rightarrow (1, 1, 1)$$

$$N_5 \rightarrow (0, 0, 0), N_6 \rightarrow (1, 1, 1)$$

Prediction = Attack

Output:

NETWORK INTRUSION DATA

Record	Packets_Per_Second	Failed_Login_Attempts	Unknown_Port_Used	\
0 N1	20	0	0	
1 N2	30	1	0	
2 N3	100	5	1	
3 N4	120	6	1	
4 N5	40	1	0	
5 N6	110	7	1	

	Admin_Access_Tried	Blacklisted_IP	Class
0	0	0	Normal
1	0	0	Normal
2	1	1	Attack
3	1	1	Attack
4	0	0	Normal
5	1	1	Attack

DISTANCE CALCULATIONS

Record	Class	Euclidean	Manhattan	Minkowski	Hamming
N1	Normal	70.1285	76	70.0045	2
N2	Normal	60.0916	65	60.0027	2
N3	Attack	10.0995	12	10.0067	1
N4	Attack	30.0832	33	30.0033	1
N5	Normal	50.1099	55	50.0039	2
N6	Attack	20.2485	24	20.0233	1

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Record	Class	Distance
N3	Attack	10.099505
N6	Attack	20.248457

N4 Attack 30.083218

Majority Voting:

Class

Attack 3

Name: count, dtype: int64

Prediction: Attack

=====

MANHATTAN DISTANCE

=====

3 Nearest Neighbours:

Record	Class	Distance
--------	-------	----------

N3	Attack	12
----	--------	----

N6	Attack	24
----	--------	----

N4	Attack	33
----	--------	----

Majority Voting:

Class

Attack 3

Name: count, dtype: int64

Prediction: Attack

=====

MINKOWSKI DISTANCE (p = 3)

=====

3 Nearest Neighbours:

Record	Class	Distance
--------	-------	----------

N3	Attack	10.006662
----	--------	-----------

N6	Attack	20.023306
----	--------	-----------

N4	Attack	30.003333
----	--------	-----------

Majority Voting:

Class

Attack 3

Name: count, dtype: int64

Prediction: Attack

=====

HAMMING DISTANCE

=====

3 Nearest Neighbours:

Record	Class	Distance
--------	-------	----------

N4	Attack	1
----	--------	---

N3	Attack	1
----	--------	---

N6	Attack	1
----	--------	---

Majority Voting:

Class

Attack 3

Name: count, dtype: int64

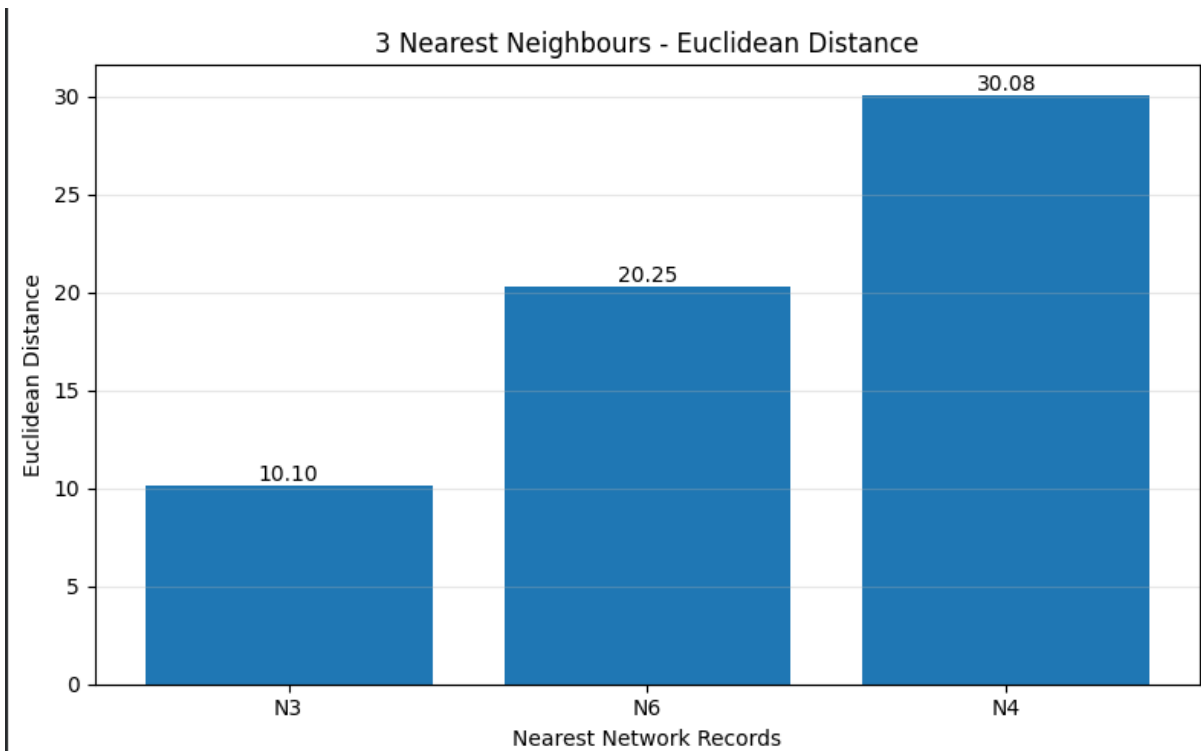
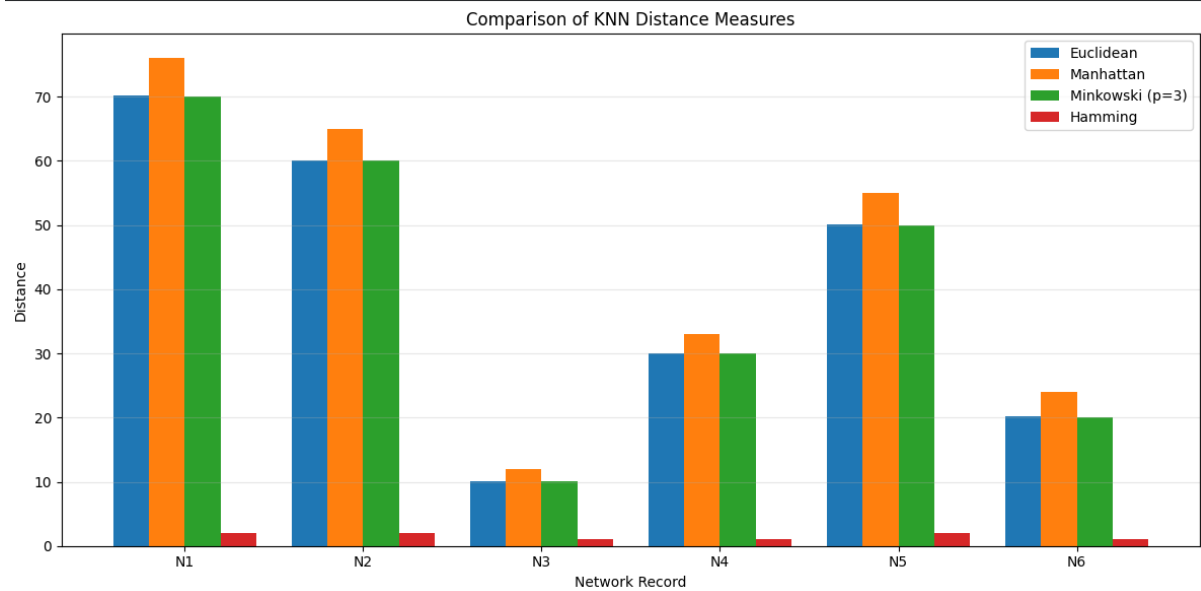
Prediction: Attack

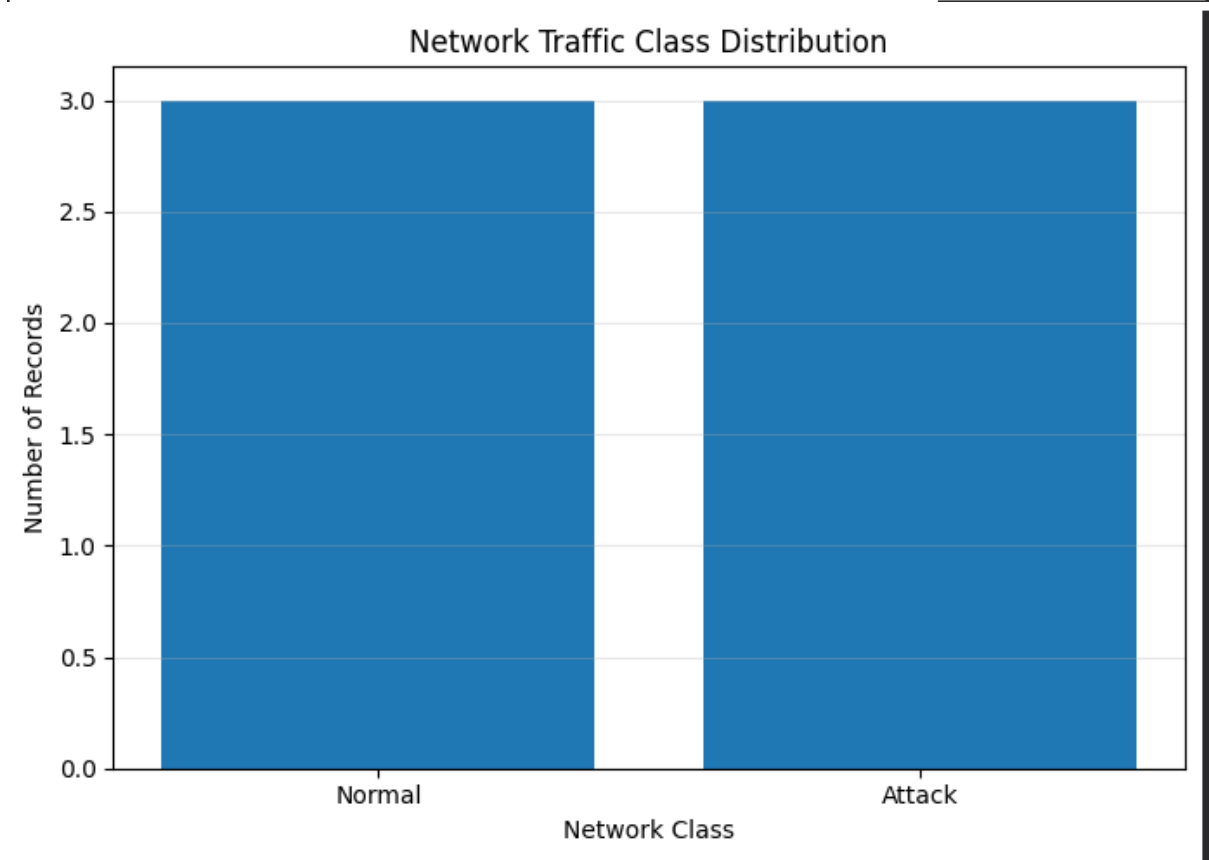
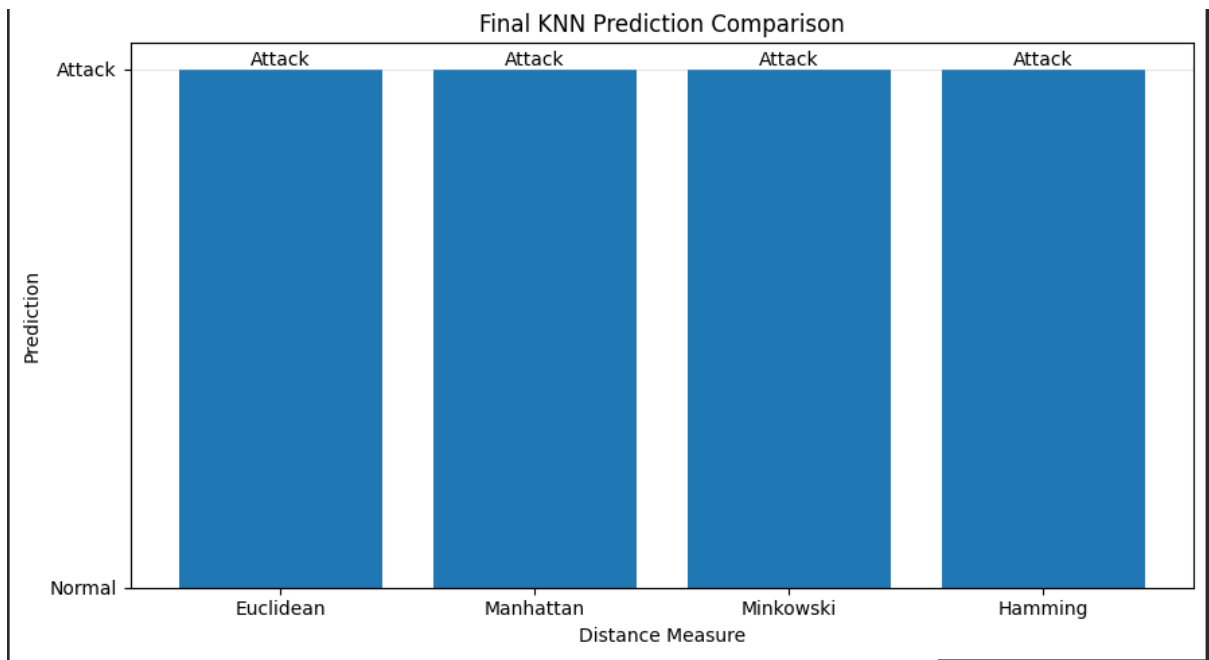
=====

FINAL PREDICTIONS

=====

Euclidean : Attack
Manhattan : Attack
Minkowski : Attack
Hamming : Attack





Question 10: Employee Attrition Prediction

A company wants to predict whether an employee is likely to Stay or Leave.

Employee	Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict	Class
E1	2	8	0	1	0	Stay
E2	3	7	0	1	0	Stay
E3	6	3	1	0	1	Leave
E4	7	2	1	0	1	Leave
E5	4	6	0	0	0	Stay
E6	8	2	1	0	1	Leave

A new employee has:

Experience in Years	Satisfaction Score	Overtime	Promotion Received	Workplace Conflict
5	4	1	0	1

Using KNN with $K = 3$, classify the employee as Stay or Leave using all four distance measures.

Question 10: Employee Attrition Prediction

$X = (5, 4, 1, 0, 1)$

$k = 3$

1. Euclidean Distance

$$D_e = \sqrt{\sum (x_i - y_i)^2}$$

$$E_1 = \sqrt{3^2 + (-4)^2 + 1^2 + (-1)^2 + 1^2} = 5.2915$$

$$E_2 = 4.0000, E_3 = 1.4142, E_4 = 2.8284, E_5 = 2.1414, E_6 = 3.6055$$

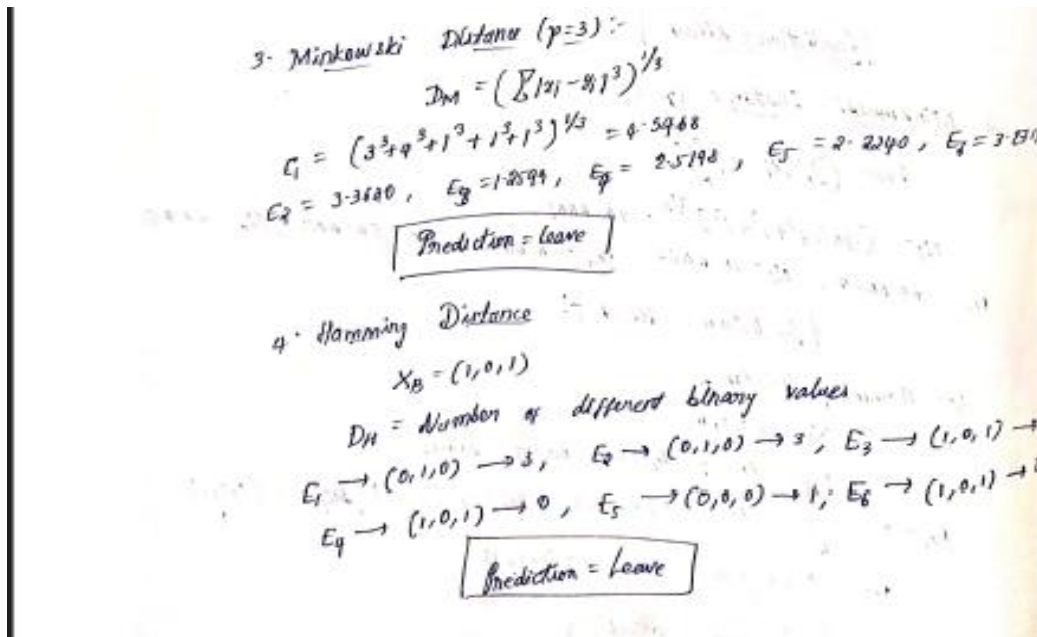
Prediction: Leave

2. Manhattan Distance

$$D_m = \sum |x_i - y_i|$$

$$E_1 = 3 + 4 + 1 + 1 + 1 = 10, E_2 = 2 + 3 + 1 + 1 + 1 = 8, E_3 = 1 + 1 + 1 + 0 + 1 = 4, E_4 = 2 + 2 + 1 + 0 + 1 = 6, E_5 = 1 + 2 + 1 + 0 + 1 = 5, E_6 = 3 + 2 + 0 + 0 + 0 = 5$$

Prediction: Leave



Output:

EMPLOYEE ATTRITION DATA

	Employee	Experience_Years	Satisfaction_Score	Overtime	\
0	E1	2	8	0	
1	E2	3	7	0	
2	E3	6	3	1	
3	E4	7	2	1	
4	E5	4	6	0	
5	E6	8	2	1	

	Promotion_Received	Workplace_Conflict	Class
0	1	0	Stay
1	1	0	Stay
2	0	1	Leave
3	0	1	Leave
4	0	0	Stay
5	0	1	Leave

DISTANCE CALCULATIONS

Employee	Class	Euclidean	Manhattan	Minkowski	Hamming
E1	Stay	5.2915	10	4.5468	3
E2	Stay	4.0000	8	3.3620	3
E3	Leave	1.4142	2	1.2599	0
E4	Leave	2.8284	4	2.5198	0
E5	Stay	2.6458	5	2.2240	2
E6	Leave	3.6056	5	3.2711	0

EUCLIDEAN DISTANCE

3 Nearest Neighbours:

Employee	Class	Distance
E3	Leave	1.414214
E5	Stay	2.645751

E4 Leave 2.828427

Majority Voting:

Class

Leave 2

Stay 1

Name: count, dtype: int64

Prediction: Leave

=====

MANHATTAN DISTANCE

=====

3 Nearest Neighbours:

Employee Class	Distance
----------------	----------

E3 Leave	2
----------	---

E4 Leave	4
----------	---

E6 Leave	5
----------	---

Majority Voting:

Class

Leave 3

Name: count, dtype: int64

Prediction: Leave

=====

MINKOWSKI DISTANCE (p=3)

=====

3 Nearest Neighbours:

Employee Class	Distance
----------------	----------

E3 Leave	1.259921
----------	----------

E5 Stay	2.223980
---------	----------

E4 Leave	2.519842
----------	----------

Majority Voting:

Class

Leave 2

Stay 1

Name: count, dtype: int64

Prediction: Leave

=====

HAMMING DISTANCE

=====

3 Nearest Neighbours:

Employee Class	Distance
----------------	----------

E4 Leave	0
----------	---

E3 Leave	0
----------	---

E6 Leave	0
----------	---

Majority Voting:

Class

Leave 3

Name: count, dtype: int64

Prediction: Leave

=====

FINAL PREDICTIONS

=====

Euclidean : Leave
Manhattan : Leave
Minkowski : Leave
Hamming : Leave

